15

Oscillatory Motion

CHAPTER OUTLINE

15.1 Motion of an Object Attached to a Spring

15.2 Analysis Model: Particle in Simple Harmonic Motion

15.3 Energy of the Simple Harmonic Oscillator

15.4 Comparing Simple Harmonic Motion with Uniform Circular Motion

15.5 The Pendulum

15.6 Damped Oscillations

15.7 Forced Oscillations

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ15.1 Answer (d). The period of a simple pendulum is $T = 2\pi\sqrt{\ell/g}$, and its frequency is $f = 1/T = (1/2\pi)\sqrt{g/\ell}$. Thus, if the length is doubled so $\ell' = 2\ell$, the new frequency is

$$f' = \frac{1}{2\pi}\sqrt{\frac{g}{\ell'}} = \frac{1}{2\pi}\sqrt{\frac{g}{2\ell}} = \frac{1}{\sqrt{2}}\left(\frac{1}{2\pi}\sqrt{\frac{g}{\ell}}\right) = \frac{f}{\sqrt{2}}$$

OQ15.2 Answer (c). The equilibrium position is 15 cm below the starting point. The motion is symmetric about the equilibrium position, so the two turning points are 30 cm apart.

OQ15.3 Answer (a). In this spring-mass system, the total energy equals the elastic potential energy at the moment the mass is temporarily at rest at $x = A = 6$ cm (i.e., at the extreme ends of the simple harmonic motion). Thus, $E = kA^2/2$ and we see that as long as the spring constant k and the amplitude A remain unchanged, the total energy is unchanged.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

OQ15.4 Answer (c). The total energy of the object-spring system is $\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$. When the kinetic energy is twice the potential energy, $\frac{1}{2}mv^2 = 2\left(\frac{1}{2}kx^2\right) = kx^2$, and the total energy is

$$\frac{1}{2}kA^2 = kx^2 + \frac{1}{2}kx^2 \quad \rightarrow \quad \frac{1}{2}kA^2 = \frac{3}{2}kx^2 \quad \rightarrow \quad x = \frac{A}{\sqrt{3}}$$

OQ15.5 Answer (d). When the object is at its maximum displacement, the magnitude of the force exerted on it by the spring is $F_s = k|x_{\max}| = (8.0\ \text{N/m})(0.10\ \text{m}) = 0.80\ \text{N}$. This force will give the mass an acceleration of $a = F_s/m = 0.80\ \text{N}/0.40\ \text{kg} = 2.0\ \text{m/s}^2$.

OQ15.6 Answer (a). The car will continue to compress the spring until all of the car's original kinetic energy has been converted into elastic potential energy within the spring, i.e., until $\frac{1}{2}kx^2 = \frac{1}{2}mv_i^2$, or

$$x = v_i\sqrt{\frac{m}{k}} = (2.0\ \text{m/s})\sqrt{\frac{3.0\times10^5\ \text{kg}}{2.0\times10^6\ \text{N/m}}} = 0.77\ \text{m}$$

OQ15.7 Answer (c). When an object undergoes simple harmonic motion, the position as a function of time may be written as $x = A\cos\omega t = A\cos(2\pi ft)$. Comparing this to the given relation, we see that the frequency of vibration is $f = 3$ Hz, and the period is $T = 1/f = 1/3$.

***OQ15.8** Answer (b). The frequency of vibration is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Thus, increasing the mass by a factor of 9 will decrease the frequency to 1/3 of its original value.

OQ15.9 Answer (a). Higher frequency. When it supports your weight, the center of the diving board flexes down less than the end does when it supports your weight—this is similar to a spring that stretches a smaller distance for the same force: its spring constant is greater because the displacement is smaller. Therefore, the stiffness constant describing the center of the board is greater than the stiffness constant describing the end. And then $f = \left(\frac{1}{2\pi}\right)\sqrt{\frac{k}{m}}$ is greater for you bouncing on the center of the board.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

OQ15.10 (i) Answer (c). At 40 cm we have the midpoint between the turning points, so it is the equilibrium position and the point of maximum speed, and therefore, maximum momentum.

(ii) Answer (c). The position of maximum speed is also the position of maximum kinetic energy.

(iii) Answer (e). The total energy of the system is conserved, so it is the same at every position.

OQ15.11 The ranking is (c) > (e) > (a) = (b) > (d). The amplitude does not affect the period in simple harmonic motion; neither do constant forces that offset the equilibrium position. Thus (a) and (b) have equal periods. The period is proportional to the square root of mass divided by spring constant. So (c), with larger mass, has a larger period than (a). And (d) with greater stiffness has smaller period. In situation (e) the motion is not quite simple harmonic, but has slightly smaller angular frequency and so a slightly longer period.

OQ15.12 (a) Yes. In simple harmonic motion, one-half of the time, the velocity is in the same direction as the displacement away from equilibrium.

(b) Yes. Velocity and acceleration are in the same direction half the time.

(c) No. The spring force and, therefore, the acceleration are always opposite to the position vector, and never in the same direction.

OQ15.13 Answer (d). We assume that the coils of the spring do not hit one another. When the spring with two blocks is set into oscillation in space, the coil in the center of the spring does not move. We can imagine clamping the center coil in place without affecting the motion. We can effectively duplicate the motion of each individual block in space by hanging a single block on a half-spring here on Earth. The half-spring with its center coil clamped—or its other half cut off—has twice the spring constant as the original uncut spring because an applied force of the same size would produce only one-half the extension distance. Thus the oscillation frequency in space is $\left(\frac{1}{2\pi}\right)\left(\frac{2k}{m}\right)^{1/2} = \sqrt{2}f$. The absence of a force required to support the vibrating system in orbital free fall has no effect on the frequency of its vibration.

OQ15.14 Answer (d) is the only false statement. At the equilibrium position, $x = 0$, the elastic potential energy of the system $\left(PE_s = \frac{1}{2}kx^2\right)$ is a minimum and the kinetic energy is a maximum.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

OQ15.15 (i) Answer (e). We have $T_i = 2\pi\sqrt{\frac{L_i}{g}}$ and $T_f = 2\pi\sqrt{\frac{L_f}{g}} = 2\pi\sqrt{\frac{4L_i}{g}} = 2T_i$. The period becomes larger by a factor of 2, to become 5 s.

(ii) Answer (c). Changing the mass has no effect on the period of a simple pendulum.

OQ15.16 (i) Answer (b). The upward acceleration has the same effect as an increased gravitational acceleration.

(ii) Answer (a). The downward acceleration has the same effect as a decreased gravitational acceleration.

(iii) Answer (c). The absence of acceleration means that the effective gravitational field is the same as that for a stationary elevator.

OQ15.17 (i) Answer (c). At 120 cm we have the midpoint between the turning points, so it is the equilibrium position and the point of maximum speed.

(ii) Answer (a). In simple harmonic motion the acceleration is maximum when the displacement from equilibrium is maximum.

(iii) Answer (a), by the same logic as in part (ii).

ANSWERS TO CONCEPTUAL QUESTIONS

CQ15.1 An imperceptibly slight breeze blowing over the edge of a leaf can produce fluttering in the same way that a breeze can cause a flag to flap. As a leaf twists in the wind, the fibers in its stem provide a restoring torque. If the frequency of the breeze matches the natural frequency of vibration of one particular leaf as a torsional pendulum, that leaf can be driven into a large-amplitude resonance vibration. Note that it is not the *size* of the driving force that sets the leaf into resonance, but the *frequency* of the driving force. If the frequency changes, another leaf will be set into resonant oscillation.

CQ15.2 (a) No. Since the acceleration is not constant in simple harmonic motion, none of the equations in Table 2.2 are valid.

(b)

Equation	Information given by equation
$x(t) = A\cos(\omega t + \phi)$	position as a function of time

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

$v(t) = -\omega A \sin(\omega t + \phi)$	velocity as a function of time
$v(x) = \pm\omega\left(A^2 - x^2\right)^{1/2}$	velocity as a function of position
$a(t) = -\omega^2 A \cos(\omega t + \phi)$	acceleration as a function of time
$a(t) = -\omega^2 x(t)$	acceleration as a function of position

(c) The angular frequency ω appears in every equation.

CQ15.3 (a) The general equation of position is $x(t) = A\cos(\omega t + \phi)$. If $x = -A\cos(\omega t)$, then $\phi = \pi$, or equally well, $\phi = -\pi$.

(b) At $t = 0$, the particle is at its turning point on the negative side of equilibrium, at $x = -A$.

CQ15.4 We assume the diameter of the bob is not very small compared to the length of the cord supporting it. As the water leaks out, the center of mass of the bob moves down, increasing the effective length of the pendulum and slightly lowering its frequency. As the last drops of water dribble out, the center of mass of the bob moves back up to the center of the sphere, and the pendulum frequency quickly increases to its original value.

CQ15.5 (a) No force is exerted on the particle. The particle moves with constant velocity.

(b) The particle feels a constant force toward the left. It moves with constant acceleration toward the left. If its initial push is toward the right, it will slow down, turn around, and speed up in motion toward the left. If its initial push is toward the left, it will just speed up.

(c) A constant force toward the right acts on the particle to produce constant acceleration toward the right.

(d) The particle moves in simple harmonic motion about the lowest point of the potential energy curve.

CQ15.6 Most everyday vibrations are damped, they eventually die down as their energy is transferred to their surroundings. However, as you will learn later, atoms in the molecules have vibration modes that do not damp out.

CQ15.7 The mechanical energy of a damped oscillator changes back and forth between kinetic and potential while it gradually and permanently decreases and transforms to internal energy.

CQ15.8 Yes. An oscillator with damping can vibrate at resonance with

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

amplitude that remains constant in time. Without damping, the amplitude would increase without limit at resonance.

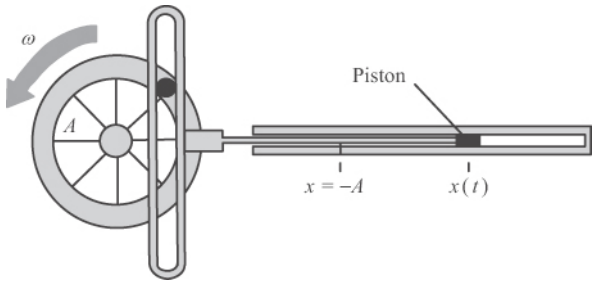
CQ15.9 No. If the resistive force is large compared to the restoring force of the spring (in particular, if $b^2 > 4mk$), the system will be overdamped and will not oscillate.

CQ15.10 The period of a pendulum depends on the acceleration of gravity: $T = 2\pi\sqrt{\frac{L}{g}}$. If the acceleration of gravity is different at the top of the mountain, the period is different and the pendulum does not keep perfect time. Two things can effect the acceleration of gravity, the top of the mountain is farther from the center of the Earth, and the nearby large mass of the mountain under the pendulum.

CQ15.11 Neither are examples of simple harmonic motion, although they are both periodic motion. In neither case is the acceleration proportional to the displacement from an equilibrium position. Neither motion is so smooth as SHM. The ball's acceleration is very large when it is in contact with the floor, and the student's when the dismissal bell rings.

CQ15.12 The motion will be periodic—that is, it will repeat, though it is not harmonic at large angles. The period is nearly constant as the angular amplitude increases through small values; then the period becomes noticeably larger as θ increases farther.

CQ15.13 The angle of the crank pin is $\theta = \omega t$. Its x coordinate is $x = A\cos\theta = A\cos\omega t$, where A is the distance from the center of the wheel to the crank pin. This is of the form $x = A\cos(\theta t + \phi)$, so the yoke and piston move with simple harmonic motion.



ANS FIG. CQ15.13

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 15.1 Motion of an Object Attached to a Spring

P15.1 (a) Taking to the right as positive, the spring force acting on the block at the instant of release is

$$F_s = -kx_i = -(130\ \text{N/m})(+0.13\ \text{m})$$
$$= -17\ \text{N} \quad \text{or} \quad \boxed{17\ \text{N to the left}}$$

(b) At this instant, the acceleration is

$$a = \frac{F_s}{m} = \frac{-17\ \text{N}}{0.60\ \text{kg}} = -28\ \text{m/s}^2$$

or $\boxed{a = 28\ \text{m/s}^2\ \text{to the left}}$

P15.2 When the object comes to equilibrium (at distance y_0 *below* the unstretched position of the end of the spring), $\sum F_y = -k(-y_0) - mg = 0$ and the force constant is

$$k = \frac{mg}{y_0} = \frac{(4.25\ \text{kg})(9.80\ \text{m/s}^2)}{2.62 \times 10^{-2}\ \text{m}} = 1.59 \times 10^3\ \text{N} = \boxed{1.59\ \text{kN/m}}$$

Section 12.2 Analysis Model: Particle in Simple Harmonic Motion

P15.3 The spring constant is found from

$$k = \frac{F_s}{x} = \frac{mg}{x} = \frac{(0.010\ \text{kg})(9.80\ \text{m/s}^2)}{3.9 \times 10^{-2}\ \text{m}} = 2.5\ \text{N/m}$$

When the object attached to the spring has mass m = 25 g, the period of oscillation is

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.025\ \text{kg}}{2.5\ \text{N/m}}} = \boxed{0.63\ \text{s}}$$

P15.4 (a) The equation for the piston's position is given as

$$x = (5.00\ \text{cm})\cos\left(2t + \frac{\pi}{6}\right)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

At $t = 0$,

$$x = (5.00\text{ cm})\cos\left(\frac{\pi}{6}\right) = \boxed{4.33\text{ cm}}$$

(b) Differentiating the equation for position with respect to time gives us the piston's velocity:

$$v = \frac{dx}{dt} = -(10.0\text{ cm/s})\sin\left(2t + \frac{\pi}{6}\right)$$

At $t = 0$, $v = \boxed{-5.00\text{ cm/s}}$

(c) Differentiating again gives its acceleration:

$$a = \frac{dv}{dt} = -\left(20.0\text{ cm/s}^2\right)\cos\left(2t + \frac{\pi}{6}\right)$$

At $t = 0$, $a = \boxed{-17.3\text{ cm/s}^2}$

(d) The period of motion is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \boxed{3.14\text{ s}}$$

(e) We read the amplitude directly from the equation for x:

$$A = \boxed{5.00\text{ cm}}$$

P15.5 $x = (4.00\text{ m})\cos(3.00\pi t + \pi)$; compare this with $x = A\cos(\omega t + \phi)$ to find

(a) $\omega = 2\pi f = 3.00\pi$ or $\boxed{f = 1.50\text{ Hz}}$

(b) $T = \frac{1}{f} = \boxed{0.667\text{ s}}$

(c) $A = \boxed{4.00\text{ m}}$

(d) $\phi = \boxed{\pi\text{ rad}}$

(e) $x(t = 0.250\text{ s}) = (4.00\text{ m})\cos(1.75\pi) = \boxed{2.83\text{ m}}$

P15.6 From the information given, we write the equation for position as $x = A\cos\omega t$, with the amplitude given as $A = 0.050\,0$ m. Differentiating gives us the piston's velocity,

$$v = -A\omega\sin\omega t$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

and differentiating again gives its acceleration

$$a = -A\omega^2 \cos \omega t$$

Then, if $f = 3600$ rev/min = 60 Hz, then $\omega = 2\pi f = 120\pi \text{ s}^{-1}$

(a) $v_{\max} = \omega A = (120\pi)(0.050\ 0) \text{ m/s} = \boxed{18.8 \text{ m/s}}$

(b) $a_{\max} = \omega^2 A = (120\pi)^2 (0.050\ 0) \text{ m/s}^2 = \boxed{7.11 \text{ km/s}^2}$

15.7 The 0.500 s must elapse between one turning point and the other. Thus the period is 1.00 s.

$$\omega = \frac{2\pi}{T} = 6.28 \text{ s}^{-1}$$

and $v_{\max} = \omega A = \left(6.28 \text{ s}^{-1}\right)(0.100 \text{ m}) = \boxed{0.628 \text{ m/s}}$.

P15.8 (a) From the information given,

$$T = \frac{12.0 \text{ s}}{5} = \boxed{2.40 \text{ s}}$$

(b) $f = \frac{1}{T} = \frac{1}{2.40} = \boxed{0.417 \text{ Hz}}$

(c) $\omega = 2\pi f = 2\pi(0.417) = \boxed{2.62 \text{ rad/s}}$

P15.9 An object hanging from a vertical spring moves with simple harmonic motion just like an object moving without friction attached to a horizontal spring. We are given the period, which is related to the frequency of motion by $T = 1/f$. Then, since $\omega = 2\pi f = \sqrt{\frac{k}{m}}$,

$$T = \frac{1}{f} = 2\pi\sqrt{\frac{m}{k}}$$

Solving for k,

$$k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2 (7.00 \text{ kg})}{(2.60 \text{ s})^2} = \boxed{40.9 \text{ N/m}}$$

***P15.10** For a simple harmonic oscillator, the maximum speed occurs at the equilibrium position and is given by Equation 15.17:

$$v_{\max} = A\sqrt{\frac{k}{m}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Thus,

$$m = \frac{kA^2}{v_{\max}^2} = \frac{(16.0 \text{ N/m})(0.200 \text{ m})^2}{(0.400 \text{ m/s})^2} = 4.00 \text{ kg}$$

and

$$F_g = mg = (4.00 \text{ kg})(9.80 \text{ m/s}^2) = \boxed{39.2 \text{ N}}$$

***P15.11** The mass of the cube is

$$m = \rho V = (2.70 \times 10^3 \text{ kg/m}^3)(0.015 \text{ m})^3 = 9.11 \times 10^{-3} \text{ kg}$$

The spring constant of the strip of steel is

$$k = \frac{F}{x} = \frac{1.43 \text{ N}}{0.027\,5 \text{ m}} = 52.0 \text{ N/m}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{52.0 \text{ N/m}}{9.11 \times 10^{-3} \text{ kg}}} = \boxed{12.0 \text{ Hz}}$$

P15.12 (a) The spring constant of this spring is

$$k = \frac{F}{x} = \frac{mg}{x} = \frac{(0.450 \text{ kg})(9.80 \text{ m/s}^2)}{0.350 \text{ m}} = 12.6 \text{ N/m}$$

We take the x axis pointing downward, so $\phi = 0$.

$$x = A\cos\omega t = (18.0 \text{ cm})\cos\left[\sqrt{\frac{12.6 \text{ N/m}}{0.450 \text{ kg}}}(84.4 \text{ s})\right]$$

$$= (18.0 \text{ cm})\cos(446.6 \text{ rad}) = \boxed{15.8 \text{ cm}}$$

(b) Now $446.6 \text{ rad} = 71 \times 2\pi + 0.497 \text{ rad}$. In each cycle the object moves 4(18) = 72 cm, so it has moved

$71(72 \text{ cm}) + (18 - 15.8) \text{ cm} = \boxed{51.1 \text{ m}}$

(c) By the same steps, $k = \dfrac{(0.440 \text{ kg})(9.80 \text{ m/s}^2)}{0.355 \text{ m}} = 12.1 \text{ N/m}$.

$$x = A\cos\sqrt{\frac{k}{m}}t = (18.0 \text{ cm})\cos\left[\sqrt{\frac{12.1 \text{ N/m}}{0.440 \text{ kg}}}(84.4 \text{ s})\right]$$

$$= (18.0 \text{ cm})\cos(443.4 \text{ rad}) = \boxed{-15.9 \text{ cm}}$$

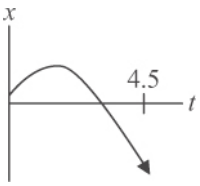
(d) $443.4 \text{ rad} = 70.569(2\pi)$

Distance moved $= 70.569(0.72 \text{ m}) = \boxed{50.8 \text{ m}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(e) The patterns of oscillation diverge from each other, starting out in phase but becoming completely out of phase. To calculate the future we would need *exact* knowledge of the present, an impossibility.

P15.13 (a) For constant acceleration position is given as a function of time by

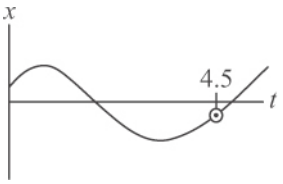


ANS. FIG. P15.13(a, b)

$$x = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$
$$= 0.270\text{ m} + (0.140\text{ m/s})(4.50\text{ s}) + \frac{1}{2}(-0.320\text{ m/s}^2)(4.50\text{ s})^2$$
$$= \boxed{-2.34\text{ m}}$$

(b) $v_x = v_{xi} + a_x t = 0.140\text{ m/s} - (0.320\text{ m/s}^2)(4.50\text{ s}) = \boxed{-1.30\text{ m/s}}$

(c) For simple harmonic motion we have instead



ANS. FIG. P15.13(c, d)

$$x = A\cos(\omega t + \phi)$$

and $v = -A\omega\sin(\omega t + \phi)$

where $a = -\omega^2 x$, so that

$-0.320\text{ m/s}^2 = -\omega^2(0.270\text{ m})$, $\omega = 1.09\text{ rad/s}$.

At $t = 0$, $0.270\text{ m} = A\cos\phi$ and $0.140\text{ m/s} = -A(1.09\text{ s}^{-1})\sin\phi$.

Dividing gives $\dfrac{0.140\text{ m/s}}{0.270\text{ m}} = -(1.09\text{ s}^{-1})\tan\phi$, $\tan\phi = -0.476$,

$\phi = -25.5°$. Still at $t = 0$, $0.270\text{ m} = A\cos(-25.5°)$, $A = 0.299\text{ m}$.

Now at $t = 4.50\text{ s}$,

$$x = (0.299\text{ m})\cos[(1.09\text{ rad/s})(4.50\text{ s}) - 25.5°]$$
$$= (0.299\text{ m})\cos(4.90\text{ rad} - 25.5°)$$
$$= (0.299\text{ m})\cos 255°$$
$$= \boxed{-0.076\,3\text{ m}}$$

(d) $v = -(0.299\text{ m})(1.09\text{ s}^{-1})\sin 255° = \boxed{+0.315\text{ m/s}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.14 (a) Since the collision is perfectly elastic, the ball will rebound to the height of 4.00 m and then repeat the motion over and over again. Thus, the $\boxed{\text{motion is periodic}}$.

(b) To determine the period, we use $x = \frac{1}{2}gt^2$. The time for the ball to hit the ground is

$$t = \sqrt{\frac{2x}{g}} = \sqrt{\frac{2(4.00\text{ m})}{9.80\text{ m/s}^2}} = 0.904\text{ s}$$

This equals one-half the period, so $T = 2(0.904\text{ s}) = \boxed{1.81\text{ s}}$.

(c) $\boxed{\text{The motion is not simple harmonic. The net force action on the ball is a constant given by } F = -mg \text{ (except when it is in contact with the ground), which is not in the form of Hooke's law.}}$

P15.15 The period of the oscillation is
$T = 1/f = 1/1.50\text{ Hz} = 1/\left(3/2\text{ s}^{-1}\right) = 2/3\text{ s}.$

(a) At $t = 0$, $x = 0$ and v is positive (to the right). Therefore, this situation corresponds to $x = A\sin\omega t$ and $v = v_i\cos\omega t$. Since $f = 1.50$ Hz, $\omega = 2\pi f = 3.00\pi$, and $A = 2.00$ cm:

$$\boxed{x = 2.00\cos\left(3.00\pi t - 90°\right) = 2.00\sin 3.00\pi t}$$

where x is in centimenters and t is in seconds.

(b) $v_{\max} = v_i = A\omega = 2.00(3.00\pi) = 6.00\pi\text{ cm/s} = \boxed{18.8\text{ cm/s}}$

(c) The particle has this speed at $t = 0$ and next after half a period:

$$t = \frac{T}{2} = \boxed{\frac{1}{3}\text{ s}}$$

(d) $a_{\max} = A\omega^2 = 2.00(3.00\pi)^2 = 18.0\pi^2\text{ cm/s}^2 = \boxed{178\text{ cm/s}^2}$

(e) This positive value of maximum acceleration first occurs when the particle is reversing its direction on the negative x axis, three-quarters of a period after $t = 0$: at $t = \frac{3}{4}T = \boxed{0.500\text{ s}}$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(f) Since $T = \frac{2}{3}$ s and $A = 2.00$ cm, the particle will travel 8.00 cm in one cycle. Hence, in $1.00 \text{ s} = \frac{3}{2}T = 1\frac{1}{2}$ cycles, the particle will travel $8.00 \text{ cm} + 4.00 \text{ cm} = \boxed{12.0 \text{ cm}}$.

***P15.16** The proposed solution,

$$x(t) = x_i \cos \omega t + \left(\frac{v_i}{\omega}\right) \sin \omega t$$

implies velocity

$$v = \frac{dx}{dt} = -x_i \omega \sin \omega t + v_i \cos \omega t$$

and acceleration

$$a = \frac{dv}{dt} = -x_i \omega^2 \cos \omega t - v_i \omega \sin \omega t$$
$$= -\omega^2 \left(x_i \cos \omega t + \left(\frac{v_i}{\omega}\right) \sin \omega t \right) = -\omega^2 x$$

(a) The acceleration being a negative constant times position means we do have SHM, and its angular frequency is ω. At $t = 0$ the equations reduce to $x = x_i$ and $v = v_i$, so they satisfy all the requirements.

(b) $v^2 - ax = v^2 - \left(-\omega^2 x\right)x = v^2 + \omega^2 x^2$

$$= \left(-x_i \omega \sin \omega t + v_i \cos \omega t\right)^2 + \omega^2 \left(x_i \cos \omega t + \left(\frac{v_i}{\omega}\right) \sin \omega t \right)^2$$
$$= x_i^2 \omega^2 \sin^2 \omega t - 2x_i v_i \omega \sin \omega t \cos \omega t + v_i^2 \cos^2 \omega t$$
$$+ x_i^2 \omega^2 \cos^2 \omega t + 2x_i v_i \omega \cos \omega t \sin \omega t$$
$$+ v_i^2 \sin^2 \omega t$$
$$= x_i^2 \omega^2 + v_i^2$$

So the expression $v^2 - ax$ is constant in time because all the parameters in the final equivalent expression $x_i^2 \omega^2 + v_i^2$ are constant. Because $v^2 - ax$ must have the same value at all times, it must be equal to the value at $t = 0$, so $v^2 - ax = v_i^2 - a_i x_i$. If we evaluate $v^2 - ax$ at a turning point where $v = 0$ and $x = A$, it is $v^2 - ax = v^2 + \omega^2 x^2 = 0^2 + \omega^2 \left(A^2\right) = \omega^2 A^2$. Thus it is proved.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

***P15.17** (a) The distance traveled in one cycle is four times the amplitude of motion, or $\boxed{20.0 \text{ cm}}$.

(b) $v_{\max} = \omega A = 2\pi fA = 2\pi(3.00 \text{ Hz})(5.00 \text{ cm}) = \boxed{94.2 \text{ cm/s}}$

This occurs as the particle passes through equilibrium.

(c) $a_{\max} = \omega^2 A = (2\pi f)^2 A = [2\pi(3.00 \text{ Hz})]^2 (0.05 \text{ m}) = \boxed{17.8 \text{ m/s}^2}$

This occurs at maximum excursion from equilibrium.

P15.18 $m = 1.00$ kg, $k = 25.0$ N/m, and $A = 3.00$ cm. At $t = 0$, $x = -3.00$ cm.

(a) $\omega = \sqrt{\dfrac{k}{m}} = \sqrt{\dfrac{25.0 \text{ N/m}}{1.00 \text{ kg}}} = 5.00 \text{ rad/s}$

so that, $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{5.00} = \boxed{1.26 \text{ s}}$

(b) $v_{\max} = A\omega = \left(3.00 \times 10^{-2} \text{ m}\right)(5.00 \text{ rad/s}) = \boxed{0.150 \text{ m/s}}$

$a_{\max} = A\omega^2 = \left(3.00 \times 10^{-2} \text{ m}\right)(5.00 \text{ rad/s})^2 = \boxed{0.750 \text{ m/s}^2}$

(c) Because $x = -3.00$ cm and $v = 0$ at $t = 0$, the required solution is $x = -A \cos \omega t$, or

$$\boxed{x = 3.00\cos(5.00t + \pi)}$$

Then, $v = \dfrac{dx}{dt} = \boxed{-15.0 \sin(5.00t + \pi)}$

and $a = \dfrac{dv}{dt} = \boxed{-75.0 \cos(5.00t + \pi)}$

where x is in cm, v is in cm/s, and a is in cm/s^2.

P15.19 $\omega = \sqrt{\dfrac{k}{m}} = \sqrt{\dfrac{8.00 \text{ N/m}}{0.500 \text{ kg}}} = 4.00 \text{ s}^{-1}$. Assuming the position of the object is at the origin at $t = 0$, position is given by $x = 10.0 \sin(4.00t)$, where x is in cm. From this, we find that $v = 40.0 \cos(4.00t)$, where v is in cm/s, and $a = -160 \sin(4.00t)$, where a is in cm/s^2.

(a) $v_{\max} = \omega A = (4.00 \text{ rad/s})(10.0 \text{ cm}) = \boxed{40.0 \text{ cm/s}}$

(b) $a_{\max} = \omega^2 A = (4.00 \text{ rad/s})^2 (10.0 \text{ cm}) = \boxed{160 \text{ cm/s}^2}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

From our assumed expression for x, we solve for the time t:

$$t = \left(\frac{1}{4.00\text{ Hz}}\right)\sin^{-1}\left(\frac{x}{10.0\text{ cm}}\right)$$

When $x = 6.00$ cm, $t = \left(\frac{1}{4.00\text{ Hz}}\right)\sin^{-1}\left(\frac{6.00\text{ cm}}{10.0\text{ cm}}\right) = 0.161$ s.

We find then that at that time:

(c) $v = (40.0\text{ cm/s})\cos\left[(4.00\text{ Hz})(0.161\text{ s})\right] = \boxed{32.0\text{ cm/s}}$ and

(d) $a = -\left(160\text{ cm/s}^2\right)\sin\left[(4.00\text{ Hz})(0.161\text{ s})\right] = \boxed{-96.0\text{ cm/s}^2}$

(e) Using $t = \left(\frac{1}{4.00\text{ Hz}}\right)\sin^{-1}\left(\frac{x}{10.0\text{ cm}}\right)$ we find that when $x = 0$, $t = 0$, and when $x = 8.00$ cm, $t = 0.232$ s. Therefore, $\Delta t = \boxed{0.232\text{ s}}$.

P15.20 (a) $\boxed{\text{Yes.}}$

(b) We assume that the mass of the spring is negligible and that we are on Earth. Let m represent the mass of the object. Its hanging at rest is described by

$$\sum F = 0 \rightarrow kx - mg = 0 \rightarrow k = \frac{mg}{x},\text{ where } x = 18.3\text{ cm}$$

To find the period, we must find the angular frequency $T = \frac{2\pi}{\omega}$.

We do not know the mass, but we do not need it because

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{mg}{x}\frac{1}{m}} = \sqrt{\frac{g}{x}}$$

From our value for x, we find

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{x}{g}} = 2\pi\sqrt{\frac{0.183\text{ m}}{9.80\text{ m/s}^2}} = 0.859\text{ s}$$

$\boxed{\text{We see that finding the period does not depend on knowing the mass: } T = 0.859\text{ s.}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Section 15.3 Energy of the Simple Harmonic Oscillator

P15.21 Choose the car with its shock-absorbing bumper as the system; by conservation of energy,

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2:$$

$$v = x\sqrt{\frac{k}{m}} = \left(3.16\times10^{-2}\text{ m}\right)\sqrt{\frac{5.00\times10^{6}\text{ N/m}}{10^{3}\text{ kg}}} = \boxed{2.23\text{ m/s}}$$

P15.22 We are given $m = 200$ g, $T = 0.250$ s, $E = 2.00$ J, and

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.250} = 25.1\text{ rad/s}$$

(a) $$k = m\omega^2 = \left(0.200\text{ kg}\right)\left(25.1\text{ rad/s}\right)^2 = \boxed{126\text{ N/m}}$$

(b) $$E = \frac{kA^2}{2} \Rightarrow A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2(2.00\text{ J})}{126\text{ N/m}}} = \boxed{0.178\text{ m}}$$

***P15.23** (a) Energy is conserved for the block-spring system between the maximum-displacement and the half-maximum points:

$$(K+U)_i = (K+U)_f$$

$$0 + \frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$\frac{1}{2}(6.50\text{ N/m})(0.100\text{ m})^2 = \frac{1}{2}m(0.300\text{ m/s})^2 + \frac{1}{2}(6.50\text{ N/m})\left(5.00\times10^{-2}\text{ m}\right)^2$$

$$3.25\times10^{-2}\text{ J} = \frac{1}{2}m(0.300\text{ m/s})^2 + 8.12\times10^{-3}\text{ J}$$

giving $$m = \frac{2\left(2.44\times10^{-2}\text{ J}\right)}{9.0\times10^{-2}\text{ m}^2/\text{s}^2} = \boxed{0.542\text{ kg}}$$

(b) $$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{6.50\text{ N/m}}{0.542\text{ kg}}} = 3.46\text{ rad/s}$$

Then, $$T = \frac{2\pi}{\omega} = \frac{2\pi\text{ rad}}{3.46\text{ rad/s}} = \boxed{1.81\text{ s}}$$

(c) $$a_{\max} = A\omega^2 = (0.100\text{ m})(3.46\text{ rad/s})^2 = \boxed{1.20\text{ m/s}^2}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

***P15.24** (a) The mechanical energy of the system is equal to the potential energy stored in the spring at maximum amplitude:

$$E = \frac{kA^2}{2} = \frac{(250\ \text{N/m})(3.50\times10^{-2}\ \text{m})^2}{2} = \boxed{0.153\ \text{J}}$$

(b) $v_{max} = A\omega$, where $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{250\ \text{N/m}}{0.500\ \text{kg}}} = 22.4\ \text{s}^{-1}$,

giving $v_{max} = \boxed{0.784\ \text{m/s}}$

(c) $a_{max} = A\omega^2 = (3.50\times10^{-2}\ \text{m})(22.4\ \text{s}^{-1})^2 = \boxed{17.5\ \text{m/s}^2}$

***P15.25** Model the oscillator as a block-spring system. From energy considerations,

$$v^2 + \omega^2 x^2 = \omega^2 A^2$$

with $v_{max} = \omega A$ and $v = \frac{\omega A}{2}$, so

$$\left(\frac{\omega A}{2}\right)^2 + \omega^2 x^2 = \omega^2 A^2$$

From this we find

$$x^2 = \frac{3}{4}A^2$$

and since A = 3.00 cm,

$$x = \pm\frac{\sqrt{3}}{2}A = \boxed{\pm 2.60\ \text{cm}}$$

***P15.26** (a) $E = \frac{1}{2}kA^2$, so if $A' = 2A$, $E' = \frac{1}{2}k(A')^2 = \frac{1}{2}k(2A)^2 = 4E$

Therefore $\boxed{E \text{ increases by factor of 4.}}$

(b) $v_{max} = \sqrt{\frac{k}{m}}A$, so if A is doubled, $\boxed{v_{max} \text{ is doubled.}}$

(c) $a_{max} = \frac{k}{m}A$, so if A is doubled, $\boxed{a_{max} \text{ also doubles.}}$

(d) $T = 2\pi\sqrt{\frac{m}{k}}$ is independent of A, so $\boxed{\text{the period is unchanged.}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.27 (a) $E = \frac{1}{2}kA^2 = \frac{1}{2}(35.0\ \text{N/m})(4.00\times10^{-2}\ \text{m})^2 = \boxed{28.0\ \text{mJ}}$

(b) $|v| = \omega\sqrt{A^2 - x^2} = \sqrt{\frac{k}{m}}\sqrt{A^2 - x^2}$

$$|v| = \sqrt{\frac{35.0\ \text{N/m}}{50.0\times10^{-3}\ \text{kg}}}\sqrt{(4.00\times10^{-2}\ \text{m})^2 - (1.00\times10^{-2}\ \text{m})^2}$$

$$= \boxed{1.02\ \text{m/s}}$$

(c) $\frac{1}{2}mv^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$

$$= \frac{1}{2}(35.0\ \text{N/m})\left[(4.00\times10^{-2}\ \text{m})^2 - (3.00\times10^{-2}\ \text{m})^2\right]$$

$$= \boxed{12.2\ \text{mJ}}$$

(d) $\frac{1}{2}kx^2 = E - \frac{1}{2}mv^2 = 28.0\ \text{mJ} - 12.2\ \text{mJ} = \boxed{15.8\ \text{mJ}}$

P15.28 (a) $k = \frac{|F|}{|x|} = \frac{20.0\ \text{N}}{0.200\ \text{m}} = \boxed{100\ \text{N/m}}$

(b) $\omega = \sqrt{\frac{k}{m}} = \sqrt{50.0}\ \text{rad/s}$ so $f = \frac{\omega}{2\pi} = \boxed{1.13\ \text{Hz}}$

(c) $v_{\max} = \omega A = \sqrt{50.0}(0.200) = \boxed{1.41\ \text{m/s}}$

(d) Maximum speed occurs when the object passes through its equilibrium position, at $\boxed{x = 0}$.

(e) $a_{\max} = \omega^2 A = 50.0(0.200) = \boxed{10.0\ \text{m/s}^2}$

(f) Maximum acceleration occurs where the object reverses direction, which is where its distance from equilibrium is a maximum, at $x = \pm A = \boxed{\pm 0.200\ \text{m}}$.

(g) $E = \frac{1}{2}kA^2 = \frac{1}{2}(100\ \text{N/m})(0.200\ \text{m})^2 = \boxed{2.00\ \text{J}}$

(h) $|v| = \omega\sqrt{A^2 - x^2} = \left(\sqrt{50.0}\ \text{rad/s}\right)\sqrt{\frac{8}{9}(0.200)^2}\ \text{m} = \boxed{1.33\ \text{m/s}}$

(i) $|a| = \omega^2 x = \left(50.0\ \text{rad}^2/\text{s}^2\right)\left(\frac{0.200\ \text{m}}{3}\right) = \boxed{3.33\ \text{m/s}^2}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.29 (a) Energy is conserved by an isolated simple harmonic oscillator:

$$E = \frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \rightarrow \frac{1}{2}mv^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

$$\rightarrow \frac{1}{2}mv^2 = \frac{1}{2}k\left(A^2 - x^2\right)$$

When $x = A/3$,

$$\frac{1}{2}mv^2 = \frac{1}{2}k\left(A^2 - x^2\right) = \frac{1}{2}k\left[A^2 - \left(\frac{A}{3}\right)^2\right] = \frac{1}{2}kA^2\left[1 - \frac{1}{9}\right]$$

$$\frac{1}{2}mv^2 = \frac{1}{2}kA^2\frac{8}{9} = \boxed{\frac{8}{9}E}$$

(b) When $x = A/3$,

$$\frac{1}{2}kx^2 = \frac{1}{2}k\left(\frac{A}{3}\right)^2 = \frac{1}{9}\left(\frac{1}{2}kA^2\right) = \boxed{\frac{1}{9}E}$$

(c) $$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}\left(\frac{1}{2}kx^2\right) + \frac{1}{2}kx^2$$

$$\frac{1}{2}kA^2 = \frac{3}{4}kx^2 \rightarrow x = \boxed{\pm\sqrt{\frac{2}{3}}A}$$

(d) No. The maximum potential energy of the system is equal to the total energy of the system: kinetic plus potential energy. Because the total energy must remain constant, the kinetic energy can never be greater than the maximum potential energy.

P15.30 (a) Particle under constant acceleration.

(b) $y_{fi} = y + v_{yi}t + \frac{1}{2}a_y t^2$:

$$-11.0\ \text{m} = 0 + 0 + \frac{1}{2}\left(-9.80\ \text{m/s}^2\right)t^2$$

$$t = \sqrt{\frac{22.0\ \text{m}}{9.80\ \text{m/s}^2}} = \boxed{1.50\ \text{s}}$$

(c) The system of the bungee jumper, the spring (cord), and the Earth is isolated.

(d) The system is isolated, so energy is conserved within the system. Take the initial point where she steps off the bridge and the final point at the bottom of her motion.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

$$\left(K + U_g + U_s\right)_i = \left(K + U_g + U_s\right)_f$$

$$0 + mgy + 0 = 0 + 0 + \frac{1}{2}kx^2$$

$$(65.0\text{ kg})(9.80\text{ m/s}^2)(36.0\text{ m}) = \frac{1}{2}k(25.0\text{ m})^2$$

which gives $k = \boxed{73.4\text{ N/m}}$

(e) The spring extension at equilibrium is

$$x = \frac{F}{k} = \frac{mg}{k} = \frac{(65.0\text{ kg})(9.80\text{ m/s}^2)}{73.4\text{ N/m}} = 8.68\text{ m}$$

so this point is $11.0 + 8.68\text{ m} = \boxed{19.7\text{ m below the bridge}}$ and the amplitude of her oscillation is 36.0 m – 19.7 m = 16.3 m.

(f) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{73.4\text{ N/m}}{65.0\text{ kg}}} = \boxed{1.06\text{ rad/s}}$

(g) Set $x = 0$ at the equilibrium position of the bungee jumper on the spring. Relative to the equilibrium position, the lowest part of the drop corresponds to $x = +16.3$ m—we have taken down as positive—and the point in the drop where the spring begins to stretch is at $x = –8.68$ m. Take the phase as zero at maximum downward extension ($x = +16.3$ m). We find that the phase, ωt, was 25 m higher where $x = –8.68$ (above the equilibrium point):

$x = A\cos\omega t$: at time $t = 0$, $x = (16.3\text{ m})\cos 0 = 16.3\text{ m}$, and when $x = –8.68 = 16.3\cos(\omega t)$, $\omega t = \pm 122° = \pm 2.13$ rad. Which sign do we pick for ωt? From $v = \frac{dx}{dt} = -\omega A\sin\omega t$, at $x = –8.68$ m, v is downward, which means by our choice of positive direction, v is positive. Pick $\omega t = –2.13$ rad:

$v = -\omega A\sin(–2.13\text{ rad}) = +\omega A(0.848)$, which is positive.

Therefore, $\omega t = 1.06t = –2.13\text{ rad} \rightarrow t = \frac{–2.13\text{ rad}}{1.06} = –2.01\text{ s}$,

meaning $t = –2.01$ s when the spring begins to stretch and $t = 0$ when the jumper reaches the bottom of the jump: then $\boxed{+2.01\text{ s}}$ is the time over which the spring stretches.

(h) total time $= 1.50\text{ s} + 2.01\text{ s} = \boxed{3.50\text{ s}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.31 (a) $F = k|x| = (83.8\ \text{N/m})(5.46\times10^{-2}\ \text{m}) = \boxed{4.58\ \text{N}}$

(b) $E = U_s = \frac{1}{2}kx^2 = \frac{1}{2}(83.8\ \text{N/m})(5.46\times10^{-2}\ \text{m})^2 = \boxed{0.125\ \text{J}}$

(c) While the block was held stationary at x = 5.46 cm, $\sum F_x = -F_s + F = 0,$ or the spring force was equal in magnitude and oppositely directed to the applied force. When the applied force is suddenly removed, there is a net force F_s = 4.58 N directed toward the equilibrium position acting on the block. This gives the block an acceleration having magnitude

$$|a| = \frac{F_s}{m} = \frac{4.58\ \text{N}}{0.250\ \text{kg}} = \boxed{18.3\ \text{m/s}^2}$$

(d) At the equilibrium position, PE_s = 0, so the block has kinetic energy $K = E$ = 0.125 J and speed

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2(0.125\ \text{J})}{0.250\ \text{kg}}} = \boxed{1.00\ \text{m/s}}$$

(e) Smaller. Friction would transform some kinetic energy into internal energy.

(f) The coefficient of kinetic friction between the block and surface.

(g) The block will come to a stop after sliding through distance $d = x$ = 0.054 6 m.

$$\Delta E_{\text{mech}} = \Delta K + \Delta U = -f_k d$$

$$0 + \left(0 - \frac{1}{2}kx^2\right) = -f_k d = -\mu_k mgd \rightarrow \mu_k = \frac{kx^2}{2mgd} = \frac{kx^2}{2mgx} = \frac{kx}{2mg}$$

$$\rightarrow \mu_k = \frac{(83.8\ \text{N/m})(0.054\ 6\ \text{m})}{2(0.250\ \text{kg})(9.80\ \text{m/s}^2)} = \boxed{0.934}$$

P15.32 (a) At the equilibrium position, the total energy of the system is in the form of kinetic energy and $mv_{\max}^2/2 = E,$ so the maximum speed is

$$v_{\max} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2(5.83\ \text{J})}{0.326\ \text{kg}}} = \boxed{5.98\ \text{m/s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) The period of an object-spring system is $T = 2\pi\sqrt{m/k}$, so the force constant of the spring is

$$k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2 (0.326 \text{ kg})}{(0.250 \text{ s})^2} = \boxed{206 \text{ N/m}}$$

(c) At the turning points, $x = \pm A$, the total energy of the system is in the form of elastic potential energy, or $E = KA^2/2$, giving the amplitude as

$$A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2(5.83 \text{ J})}{206 \text{ N/m}}} = \boxed{0.238 \text{ m}}$$

Section 15.4 Comparing Simple Harmonic Motion with Uniform Circular Motion

P15.33 (a) The motion is simple harmonic because the tire is rotating with constant angular velocity and you see the projection of the motion of the bump in a plane perpendicular to the tire.

(b) Since the car is moving with speed $v = 3.00$ m/s, and its radius is 0.300 m, we have

$$\omega = \frac{3.00 \text{ m/s}}{0.300 \text{ m}} = 10.0 \text{ rad/s}$$

Therefore, the period of the motion is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{(10.0 \text{ rad/s})} = \boxed{0.628 \text{ s}}$$

Section 15.5 The Pendulum

P15.34 The period in Tokyo is $T_T = 2\pi\sqrt{\frac{L_T}{g_T}}$, and the period in Cambridge is

$$T_C = 2\pi\sqrt{\frac{L_C}{g_C}}.$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

We know that $T_T = T_C = 2.00 \text{ s}$, which means that $\frac{L_T}{g_T} = \frac{L_C}{g_C}$,

or $$\frac{g_C}{g_T} = \frac{L_C}{L_T} = \frac{0.994\ 2}{0.992\ 7} = \boxed{1.001\ 5}$$

P15.35 The period of a pendulum is the time for one complete oscillation and is given by $T = 2\pi\sqrt{\ell/g}$, where ℓ is the length of the pendulum.

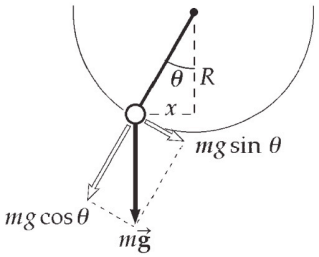
(a) $$T = \left(\frac{3.00 \text{ min}}{120 \text{ oscillations}}\right)\left(\frac{60 \text{ s}}{1 \text{ min}}\right) = \boxed{1.50 \text{ s}}$$

(b) The length of the pendulum is

$$\ell = g\left(\frac{T^2}{4\pi^2}\right) = \left(9.80 \text{ m/s}^2\right)\left(\frac{(1.50 \text{ s})^2}{4\pi^2}\right) = \boxed{0.559 \text{ m}}$$

P15.36 Referring to ANS. FIG. P15.36, we have

$$F = -mg\sin\theta \quad \text{and} \quad \tan\theta = \frac{x}{R}$$



ANS. FIG. P15.36

For small displacements,

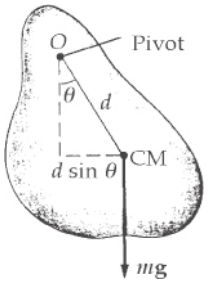
$$\tan\theta \approx \sin\theta \quad \text{and} \quad F = -\frac{mg}{R}x = -kx$$

Since the restoring force is proportional to the displacement from equilibrium, the motion is simple harmonic motion.

Comparing to $F = -m\omega^2 x$ shows

$$\boxed{\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{R}}}$$

P15.37 $f = 0.450$ Hz, $d = 0.350$ m, and $m = 2.20$ kg. Now,



ANS. FIG. P15.37

$$T = \frac{1}{f}$$

$$T = 2\pi\sqrt{\frac{I}{mgd}} \rightarrow T^2 = \frac{4\pi^2 I}{mgd}$$

Solving for the moment of inertia, we obtain

$$I = T^2\frac{mgd}{4\pi^2} = \left(\frac{1}{f}\right)^2\frac{mgd}{4\pi^2} = \frac{(2.20 \text{ kg})(9.80 \text{ m/s}^2)(0.350 \text{ m})}{4\pi^2\left(0.450 \text{ s}^{-1}\right)^2}$$

$$= \boxed{0.944 \text{ kg} \cdot \text{m}^2}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.38 Please see ANS. FIG. P15.37. For a physical pendulum,

$$T = \frac{1}{f}$$

$$T = 2\pi\sqrt{\frac{I}{mgd}} \rightarrow T^2 = \frac{4\pi^2 I}{mgd}$$

$$\rightarrow I = T^2\frac{mgd}{4\pi^2} = \left(\frac{1}{f}\right)^2\frac{mgd}{4\pi^2} \rightarrow \boxed{I = \frac{mgd}{4\pi^2 f^2}}$$

***P15.39** We solve $\omega = \frac{2\pi}{T}$ for $T = \frac{2\pi}{\omega} = \frac{2\pi}{4.43} = \boxed{1.42\text{ s}}$

We then solve $\omega = \sqrt{\frac{g}{L}}$ for $L = \frac{g}{\omega^2} = \frac{9.80\text{ m/s}^2}{(4.43\text{ rad/s})^2} = \boxed{0.499}$

P15.40 (a) The parallel-axis theorem gives $I = I_{CM} + md^2$,

so $T = 2\pi\sqrt{\frac{I}{mgd}} = \boxed{2\pi\sqrt{\frac{I_{CM} + md^2}{mgd}}}$

(b) When d is very large $T \rightarrow 2\pi\sqrt{\frac{d}{g}}$ gets large.

When d is very small $T \rightarrow 2\pi\sqrt{\frac{I_{CM}}{mgd}}$ gets large.

So there must be a minimum, found by

$$\frac{dT}{dd} = 0 = \frac{d}{dd}2\pi\left(I_{CM} + md^2\right)^{1/2}\left(mgd\right)^{-1/2}$$

$$= 2\pi\left(I_{CM} + md^2\right)^{1/2}\left(-\frac{1}{2}\right)\left(mgd\right)^{-3/2}mg$$

$$+ 2\pi\left(mgd\right)^{-1/2}\left(\frac{1}{2}\right)\left(I_{CM} + md^2\right)^{-1/2}2md$$

$$= \frac{-\pi\left(I_{CM} + md^2\right)mg}{\left(I_{CM} + md^2\right)^{1/2}\left(mgd\right)^{3/2}} + \frac{2\pi\, md\, mgd}{\left(I_{CM} + md^2\right)^{1/2}\left(mgd\right)^{3/2}} = 0$$

This requires

$$-I_{CM} - md^2 + 2md^2 = 0$$

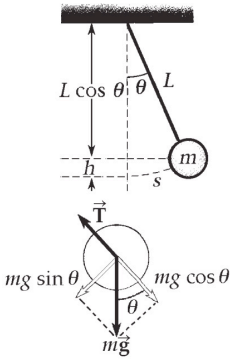
or $\boxed{I_{CM} = md^2}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.41 Using the simple harmonic motion model:

$$A = r\theta = (1.00\text{ m})\left[(15.0°)\frac{\pi}{180°}\right] = 0.262\text{ m}$$

$$\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.80\text{ m/s}^2}{1.00\text{ m}}} = 3.13\text{ rad/s}$$



ANS. FIG. P15.41

(a) $v_{max} = A\omega = (0.262\text{ m})(3.13\text{ s}^{-1})$

$= \boxed{0.820\text{ m/s}}$

(b) For simple harmonic motion, the maximum acceleration

$$a_{max} = A\omega^2 = (0.262\text{ m})(3.13\text{ s}^{-1})^2$$
$$= 2.57\text{ m/s}^2$$

which is equal to the maximum tangential acceleration, occurs at the extreme ends of the swing:

$$a_t = r\alpha \rightarrow \alpha = \frac{a_t}{r} = \frac{2.57\text{ m/s}^2}{1.00\text{ m}} = \boxed{2.57\text{ rad/s}^2}$$

(c) The maximum restoring force causes the maximum acceleration:

$$F = ma_{max} = 0.25\text{ kg}\ (2.57\text{ m/s}^2) = \boxed{0.641\text{ N}}$$

(d) (a) Applying energy conservation to the isolated pendulum-Earth system:

$$K_i + U_i = K_f + U_f \rightarrow mgh = \frac{1}{2}mv^2$$

and $h = L(1 - \cos\theta)$,

then

$$v_{max} = \sqrt{2gh} = \sqrt{2gL(1 - \cos\theta)}$$
$$= \sqrt{2(9.80\text{ m/s}^2)(1.00\text{ m})(1 - \cos 15.0°)}$$
$$= \boxed{0.817\text{ m/s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.42 (a) The parallel-axis theorem gives:

$$I = I_{CM} + Md^2 = \frac{1}{12}ML^2 + Md^2$$

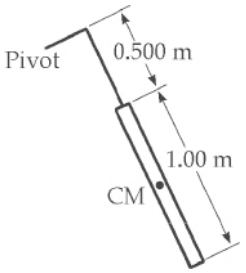
$$= \frac{1}{12}M(1.00\text{ m})^2 + M(1.00\text{ m})^2$$

$$= M\left(\frac{13}{12}\text{ m}^2\right)$$

$$T = 2\pi\sqrt{\frac{I}{Mgd}} = 2\pi\sqrt{\frac{M\left(13/12\text{ m}^2\right)}{Mg(1.00\text{ m})}}$$

$$= 2\pi\sqrt{\frac{13/12\text{ m}}{9.80\text{ m/s}^2}}$$

$$= \boxed{2.09\text{ s}}$$



ANS. FIG. P15.42

(b) For the simple pendulum,

$$T = 2\pi\sqrt{\frac{1.00\text{ m}}{9.80\text{ m/s}^2}} = 2.01\text{ s}$$

$$\text{difference} = \frac{2.09\text{ s} - 2.01\text{ s}}{2.01\text{ s}} = \boxed{4.08\%}$$

P15.43 (a) The string tension must support the weight of the bob, accelerate it upward, and also provide the restoring force, just as if the elevator were at rest in a gravity field of (9.80 + 5.00) m/s^2. Thus the period is

$$T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{5.00\text{ m}}{14.8\text{ m/s}^2}}$$

$$T = \boxed{3.65\text{ s}}$$

(b) $$T = 2\pi\sqrt{\frac{5.00\text{ m}}{\left(9.80\text{ m/s}^2 - 5.00\text{ m/s}^2\right)}} = \boxed{6.41\text{ s}}$$

(c) $$g_{\text{eff}} = \sqrt{\left(9.80\text{ m/s}^2\right)^2 + \left(5.00\text{ m/s}^2\right)^2} = 11.0\text{ m/s}^2$$

$$T = 2\pi\sqrt{\frac{5.00\text{ m}}{11.0\text{ m/s}^2}} = \boxed{4.24\text{ s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.44 (a) From $T = \dfrac{\text{total measured time}}{50}$,

the measured periods are:

Length, L (m)	1.000	0.750	0.500
Period, T (s)	2.00	1.73	1.42

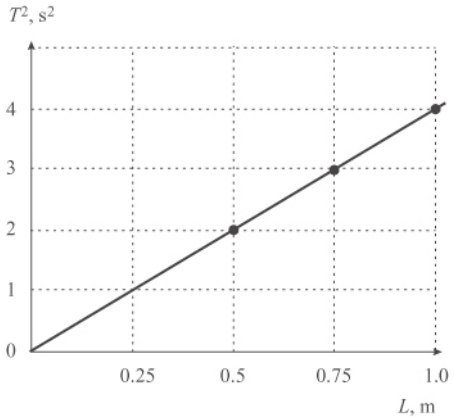
(b) $T = 2\pi\sqrt{\dfrac{L}{g}}$ so $g = \dfrac{4\pi^2 L}{T^2}$

The calculated values for g are:

Period, T (s)	2.00	1.73	1.42
g (m/s^2)	9.87	9.89	9.79

Thus, $g_{\text{avg}} = 9.85 \text{ m/s}^2$

This agrees with the accepted value of $g = 9.80 \text{ m/s}^2$ within 0.5%.



ANS. FIG. P15.44

(c) From $T^2 = \left(\dfrac{4\pi^2}{g}\right)L$, the slope of T^2 versus L graph is

$$\frac{4\pi^2}{g} = 3.97 \text{ s}^2/\text{m}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Thus, $g = \dfrac{4\pi^2}{\text{slope}} = \boxed{9.94\ \text{m/s}^2}$. This is within 1.5% of the accepted value for g.

P15.45 The period of oscillation for the watch balance wheel is $T = 0.250$ s. Modeling the 20.0-g mass as a particle, we find the moment of inertia from $I = mr^2$.

(a) $I = mr^2 = \left(2.00\times10^{-2}\ \text{kg}\right)\left(5.00\times10^{-3}\ \text{m}\right)^2 = \boxed{5.00\times10^{-7}\ \text{kg}\cdot\text{m}^2}$

(b) To find the torsion constant, we use Equation 15.29 for the motion of a torsional pendulum,

$$I\frac{d^2\theta}{dt^2} = -\kappa\theta$$

where

$$\sqrt{\frac{\kappa}{I}} = \omega = \frac{2\pi}{T}$$

Solving for the torsion constant gives

$$\kappa = I\omega^2 = \left(5.00\times10^{-7}\ \text{kg}\cdot\text{m}^2\right)\left(\frac{2\pi}{0.250\ \text{s}}\right)^2 = \boxed{3.16\times10^{-4}\ \frac{\text{N}\cdot\text{m}}{\text{rad}}}$$

Section 15.6 Damped Oscillations

P15.46 We are given $\theta_i = 15.0°$, and $\theta(t = 1\ 000\ \text{s}) = 5.50°$. We then use Equation 15.32 for damped oscillations:

$$x = Ae^{-bt/2m}$$

Substituting,

$$\frac{x_{1\,000}}{x_i} = \frac{Ae^{-bt/2m}}{A} = \frac{5.50}{15.0} = e^{-b(1\,000)/2m}$$

$$\ln\left(\frac{5.50}{15.0}\right) = -1.00 = \frac{-b(1\ 000)}{2m}$$

which gives

$$\frac{b}{2m} = \boxed{1.00\times10^{-3}\ \text{s}^{-1}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.47 If the oscillation was undamped, its frequency would be

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{2.05\times10^4\ \text{N/m}}{10.6\ \text{kg}}} = 44.0\ \text{s}^{-1}$$

(a) With damping, the frequency becomes

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} = \sqrt{\left(44\ \frac{1}{\text{s}}\right)^2 - \left(\frac{3\ \text{kg}}{2\ \times 10.6\ \text{kg}}\right)}$$

$$= \sqrt{1\,933.96 - 0.02} = 44.0\ \text{s}^{-1}$$

$$f = \frac{\omega}{2\pi} = \frac{44.0\ \text{s}^{-1}}{2\pi} = \boxed{7.00\ \text{Hz}}$$

(b) In $x = A_0 e^{-bt/2m}\cos(\omega t + \phi)$ over one cycle, a time $T = \dfrac{2\pi}{\omega}$, the amplitude changes from A_0 to $A_0 e^{-b2\pi/2m\omega}$ for a fractional decrease of

$$\frac{A_0 - A_0 e^{-\pi b/m\omega}}{A_0} = 1 - e^{-\pi 3/(10.6\cdot 44.0)} = 1 - e^{-0.020\,2} = 1 - 0.979\,98$$

$$= 0.020\ 0 = \boxed{2.00\%}$$

(c) The energy is proportional to the square of the amplitude, so its fractional rate of decrease is twice as fast:

$$E = \frac{1}{2}kA^2 = \frac{1}{2}kA_0^2 e^{-2bt/2m} = E_0 e^{-bt/m}$$

We specify

$$(0.050\ 0)E_0 = E_0 e^{-3t/10.6}$$

$$0.050\ 0 = e^{-3t/10.6}$$

$$e^{+3t/10.6} = 20.0$$

$$\frac{3t}{10.6} = \ln 20.0 = 3.00$$

$$t = \boxed{10.6\ \text{s}}$$

P15.48 The total energy is $E = \dfrac{1}{2}mv^2 + \dfrac{1}{2}kx^2$.

Taking the time derivative gives $\dfrac{dE}{dt} = mv\dfrac{d^2x}{dt^2} + kxv$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Then, substituting from Equation 15.31, $\frac{md^2x}{dt^2} = -kx - bv$, gives

$$\frac{dE}{dt} = v(-kx - bv) + kvx$$

Thus, $\boxed{\frac{dE}{dt} = -bv^2 < 0}$

We have proved that the mechanical energy of a damped oscillator is always decreasing.

P15.49 To show that $x = Ae^{-bt/2m}\cos(\omega t + \phi)$

is a solution of $-kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$ **[1]**

where $\omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$ and $b^2 < 4mk$ so that ω is real, **[2]**

we take $x = Ae^{-bt/2m}\cos(\omega t + \phi)$ and compute **[3]**

$$\frac{dx}{dt} = Ae^{-bt/2m}\left(-\frac{b}{2m}\right)\cos(\omega t + \phi) - Ae^{-bt/2m}\omega\sin(\omega t + \phi) \quad \textbf{[4]}$$

$$\frac{d^2x}{dt^2} = -\frac{b}{2m}\left[Ae^{-bt/2m}\left(-\frac{b}{2m}\right)\cos(\omega t + \phi) - Ae^{-bt/2m}\omega\sin(\omega t + \phi)\right]$$

$$-\left[Ae^{-bt/2m}\left(-\frac{b}{2m}\right)\omega\sin(\omega t + \phi) + Ae^{-bt/2m}\omega^2\cos(\omega t + \phi)\right] \quad \textbf{[5]}$$

We substitute [3] and [4] into the left side of [1], and [5] into the right side of [1]:

$$-kAe^{-bt/2m}\cos(\omega t + \phi) + \frac{b^2}{2m}Ae^{-bt/2m}\cos(\omega t + \phi) + b\omega Ae^{-bt/2m}\sin(\omega t + \phi)$$

$$= -\frac{b}{2}\left[Ae^{-bt/2m}\left(-\frac{b}{2m}\right)\cos(\omega t + \phi) - Ae^{-bt/2m}\omega\sin(\omega t + \phi)\right]$$

$$+\frac{b}{2}Ae^{-bt/2m}\omega\sin(\omega t + \phi) - m\omega^2Ae^{-bt/2m}\cos(\omega t + \phi)$$

We then compare the coefficients of the $Ae^{-bt/2m}\cos(\omega t + \phi)$ and the $Ae^{-bt/2m}\sin(\omega t + \phi)$ terms.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The cosine term is

$$-k+\frac{b^2}{2m}=-\frac{b}{2}\left(-\frac{b}{2m}\right)-m\omega^2=\frac{b^2}{4m}-m\left(\frac{k}{m}-\frac{b^2}{4m^2}\right)=-k+\frac{b^2}{2m}$$

and the sine term is

$$b\omega=+\frac{b}{2}(\omega)+\frac{b}{2}(\omega)=b\omega$$

Since the coefficients are equal, $x=Ae^{-bt/2m}\cos(\omega t+\phi)$ is a solution of the equation.

Section 15.7 Forced Oscillations

P15.50 (a) For resonance, her frequency must match:

$$f_0=\frac{\omega_0}{2\pi}=\frac{1}{2\pi}\sqrt{\frac{k}{m}}=\frac{1}{2\pi}\sqrt{\frac{7.00\times10^2\ \text{N/m}}{12.5\ \text{kg}}}=\boxed{1.19\ \text{Hz}}$$

(b) From $x=A\cos\omega t$, $v=\frac{dx}{dt}=-A\omega\sin\omega t$, and $a=\frac{dv}{dt}=-A\omega^2\cos\omega t$, the maximum acceleration is $A\omega^2$. When this becomes equal to the acceleration due to gravity, the normal force exerted on her by the mattress will drop to zero at one point in the cycle:

$$A\omega^2=g \quad \text{or} \quad A=\frac{g}{\omega^2}=\frac{g}{k/m}=\frac{gm}{k}$$

$$A=\frac{(9.80\ \text{m/s}^2)(12.5\ \text{kg})}{7.00\times10^2\ \text{N/m}}=\boxed{17.5\ \text{cm}}$$

P15.51 The pendulum is resonating with the beeper. The beeper must vibrate at the frequency of a simple pendulum of frequency 1.50 Hz:

$$\omega=2\pi f=\sqrt{\frac{g}{L}}\rightarrow L=\frac{g}{(2\pi f)^2}=\frac{9.80\ \text{m/s}^2}{[2\pi(1.50\ \text{Hz})]^2}$$

$$=0.110\ \text{m}=\boxed{11.0\ \text{cm}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.52 From the equation for the amplitude of a driven oscillator with no damping,

$$A = \frac{F_0/m}{\sqrt{\left(\omega^2 - \omega_0^2\right)^2}} = \frac{F_0/m}{\omega^2 - \omega_0^2}$$

which gives

$$F_0 = mA\left(\omega^2 - \omega_0^2\right)$$

The driving frequency is

$$\omega^2 = \left(2\pi f\right)^2 = \left[2\pi\left(10.0 \text{ s}^{-1}\right)\right]^2 = 3.95\times10^3 \text{ s}^{-2}$$

and the natural frequency of the oscillator is

$$\omega_0^2 = \frac{k}{m} = \frac{200 \text{ N/m}}{40.0 \text{ N} / 9.80 \text{ m/s}^2} = 49.0 \text{ s}^{-2}$$

Substituting gives us a driving force of

$$F_0 = \left(\frac{40.0 \text{ N}}{9.80 \text{ m/s}^2}\right)\left(2.00\times10^{-2} \text{ m}\right)\left(3.95\times10^3 \text{ s}^{-2} - 49.0 \text{ s}^{-2}\right)$$
$$= \boxed{318 \text{ N}}$$

P15.53 We are given $F = 3.00 \sin(2\pi t)$, $k = 20.0$ N/m, and $m = 2.00$ kg.

(a) $$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{20.0 \text{ N/m}}{2.00 \text{ kg}}} = \boxed{3.16 \text{ s}^{-1}}$$

(b) From $F = 3.00 \sin(2\pi t)$, the angular frequency of the force is

$$\omega = 2\pi = \boxed{6.28 \text{ s}^{-1}}$$

(c) From equation 15.36, the amplitude A of a driven oscillator, with $b = 0$, gives

$$A = \frac{F_0 / m}{\omega^2 - \omega_0^2} = \frac{\left(3.00 \text{ N/m}\right)/\left(2.00 \text{ kg}\right)}{\left(6.28 \text{ s}^{-1}\right)^2 - \left(3.16 \text{ s}^{-1}\right)^2} = 0.050\ 9 \text{ m} = \boxed{5.09 \text{ cm}}$$

P15.54 We start with Equation 15.34, $F_0 \sin \omega t - kx = m\dfrac{d^2x}{dt^2}$ **[1]**

Equation 15.35 gives the solution to this equation as

$$x = A \cos\left(\omega t + \phi\right) \quad \textbf{[2]}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

differentiating,

$$\frac{dx}{dt} = -A\omega \sin(\omega t + \phi) \qquad \textbf{[3]}$$

and differentiating again,

$$\frac{d^2x}{dt^2} = -A\omega^2 \cos(\omega t + \phi) \qquad \textbf{[4]}$$

Substituting [2] and [4] into [1]:

$$F_0 \sin \omega t - kA \cos(\omega t + \phi) = m\left(-A\omega^2\right) \cos(\omega t + \phi)$$

Solving for the amplitude:

$$\left(kA - mA\omega^2\right) \cos(\omega t + \phi) = F_0 \sin \omega t = -F_0 \cos(\omega t + 90°)$$

These will be equal, provided only that ϕ must be +90° and

$$kA - mA\omega^2 = -F_0$$

Thus, $A = \dfrac{F_0/m}{\omega^2 - \omega_0^2}$, where $\omega_0 = \sqrt{\dfrac{k}{m}}$.

P15.55 We use the equation for the amplitude of forced oscillations,

$$A = \frac{F_{\text{ext}}/m}{\sqrt{\left(\omega^2 - \omega_0^2\right)^2 + \left(b\omega/m\right)^2}}$$

With $b = 0$,

$$A = \frac{F_{\text{ext}}/m}{\sqrt{\left(\omega^2 - \omega_0^2\right)^2}} = \frac{F_{\text{ext}}/m}{\pm\left(\omega^2 - \omega_0^2\right)} = \pm\frac{F_{\text{ext}}/m}{\omega^2 - \omega_0^2}$$

Thus,

$$\omega^2 = \omega_0^2 \pm \frac{F_{\text{ext}}/m}{A} = \frac{k}{m} \pm \frac{F_{\text{ext}}}{mA}$$
$$= \frac{6.30 \text{ N/m}}{0.150 \text{ kg}} \pm \frac{1.70 \text{ N}}{(0.150 \text{ kg})(0.440 \text{ m})}$$

This yields $\omega = 8.23$ rad/s or $\omega = 4.03$ rad/s. Then,

$$f = \frac{\omega}{2\pi} \text{ gives either } f = \boxed{1.31 \text{ Hz}} \quad \text{or} \quad f = \boxed{0.641 \text{ Hz}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Additional Problems

P15.56 Deuterium is the isotope of the element hydrogen with atoms having nuclei consisting of one proton and one neutron. For brevity we refer to the molecule formed by two deuterium atoms as *D* and to the diatomic molecule of hydrogen-1 as *H*, with $M_D = 2M_H$.

$$\frac{\omega_D}{\omega_H} = \frac{\sqrt{k/M_D}}{\sqrt{k/M_H}} = \sqrt{\frac{M_H}{M_D}} = \sqrt{\frac{1}{2}}$$

$$f_D = \frac{f_H}{\sqrt{2}} = \frac{1.30 \times 10^{14}\ \text{Hz}}{\sqrt{2}} = \boxed{0.919 \times 10^{14}\ \text{Hz}}$$

P15.57 From $a = -\omega^2 x$, the maximum acceleration is given by $a_{\max} = \omega^2 A$. Then $108\ \text{cm/s}^2 = \omega^2\ (12.0\ \text{cm})$, giving $\omega = 3.00\ \text{rad/s}$.

(a) $T = 1/f = 2\pi/\omega = 2\pi/\left(3.00\ \text{s}^{-1}\right) = \boxed{2.09\ \text{s}}$

(b) $f = \omega/2\pi = \left(3.00\ \text{s}^{-1}\right)/2\pi = \boxed{0.477\ \text{Hz}}$

(c) $v_{\max} = \omega A = \left(3\ \text{s}^{-1}\right)\left(12.0\ \text{cm}\right) = \boxed{36.0\ \text{cm/s}}$

(d) $E = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} m (0.360\ \text{m/s})^2$

$= \boxed{0.064\ 8m,\ \text{where } E \text{ is in joules and } m \text{ is in kg}}$

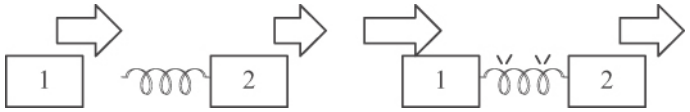
(e) From $\omega^2 = \frac{k}{m}$, we obtain

$$k = \omega^2 m = \left(3.00\ \text{s}^{-1}\right)^2 m$$

$$= \boxed{9.00m,\ \text{where } k \text{ is in newtons/meter and } m \text{ is in kg}}$$

(f) Period, frequency, and maximum speed are all independent of mass in this situation. The energy and the force constant are directly proportional to mass.

P15.58 (a) Consider the first process of spring compression. It continues as long as glider 1 is moving faster than glider 2. The spring instantaneously has maximum compression when both gliders are moving with the same speed v_a.



ANS. FIG. P15.58(a)

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Momentum conservation then gives

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$(0.240\ \text{kg})(0.740\ \text{m/s}) + (0.360\ \text{kg})(0.12\ \text{m/s}) = (0.240\ \text{kg})v_a + (0.360\ \text{kg})v_a$$

$$\frac{0.220\ 8\ \text{kg}\cdot\text{m/s}}{0.600\ \text{kg}} = v_a$$

$$v_a = \boxed{0.368\ \text{m/s}}$$

(b) From energy conservation, we have

$$(K_1 + K_2 + U_s)_i = (K_1 + K_2 + U_s)_f$$

$$\frac{1}{2}m_1 v_{1i}^2 + \frac{1}{2}m_2 v_{2i}^2 + 0 = \frac{1}{2}(m_1 + m_2)v_a^2 + \frac{1}{2}kx^2$$

$$\frac{1}{2}(0.240\ \text{kg})(0.740\ \text{m/s})^2 + \frac{1}{2}(0.360\ \text{kg})(0.120\ \text{m/s})^2 = \frac{1}{2}(0.600\ \text{kg})(0.368\ \text{m/s})^2 + \frac{1}{2}(45.0\ \text{N/m})x^2$$

$$0.068\ 3\ \text{J} = 0.040\ 6\ \text{J} + \frac{1}{2}(45.0\ \text{N/m})x^2$$

$$x = \left(\frac{2(0.027\ 7\ \text{J})}{45.0\ \text{N/m}}\right)^{1/2} = 0.035\ 1\ \text{m} = \boxed{3.51\ \text{cm}}$$

(c) $$\frac{1}{2}m_{\text{tot}}v_{\text{CM}}^2 = \frac{1}{2}(0.600\ \text{kg})(0.368\ \text{m/s})^2 = 0.040\ 6\ \text{J} = \boxed{40.6\ \text{mJ}}$$

(d) $$\frac{1}{2}kx^2 = \frac{1}{2}(45.0\ \text{N/m})(0.0351\ \text{m})^2 = 0.027\ 7\ \text{J} = \boxed{27.7\ \text{mJ}}$$

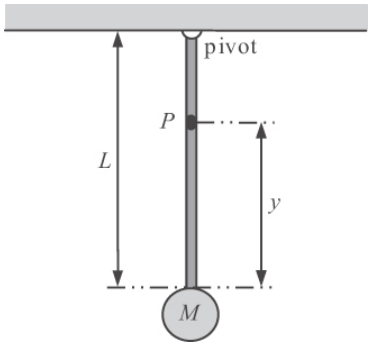
P15.59 Let F represent the tension in the rod.

(a) At the pivot,

$$F = Mg + Mg = \boxed{2Mg}$$

(b) A fraction of the rod's weight $Mg\left(\frac{y}{L}\right)$ as well as the weight of the ball pulls down on point P. Thus, the tension in the rod at point P is

$$F = Mg\left(\frac{y}{L}\right) + Mg = \boxed{Mg\left(1 + \frac{y}{L}\right)}$$



ANS. FIG. P15.59

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) Relative to the pivot, $I = I_{\text{rod}} + I_{\text{ball}} = \frac{1}{3}ML^2 + ML^2 = \frac{4}{3}ML^2$.

For the physical pendulum, $T = 2\pi\sqrt{\frac{I}{mgd}}$, where $m = 2M$ and d is the distance from the pivot to the center of mass of the rod and ball combination. Therefore,

$$d = \frac{M(L/2) + ML}{M + M} = \frac{3L}{4}$$

and $T = 2\pi\sqrt{\frac{(4/3)ML^2}{(2M)g(3L/4)}} = \boxed{\frac{4\pi}{3}\sqrt{\frac{2L}{g}}}$.

(d) For $L = 2.00$ m, $T = \frac{4\pi}{3}\sqrt{\frac{2(2.00\text{ m})}{9.80\text{ m/s}^2}} = \boxed{2.68\text{ s}}$.

P15.60 (a) From $a = -\omega^2 x$, the maximum acceleration is given by $a_{\max} = \omega^2 A$. As A increases, the maximum acceleration increases. When it becomes greater than the free-fall acceleration, the rock will no longer stay in contact with the vibrating ground, but lag behind as the ground moves down with greater acceleration. We have then

$$A = \frac{g}{\omega^2} = \frac{g}{(2\pi f)^2} = \frac{9.80\text{ m/s}^2}{\left[2\pi(2.40\text{ s}^{-1})\right]^2} = \boxed{4.31\text{ cm}}$$

(b) When the rock is on the point of lifting off, the surrounding water is also barely in free fall. No pressure gradient exists in the water, so no buoyant force acts on the rock. The effect of the surrounding water disappears at that instant.

***P15.61** For the resonance vibration with the occupants in the car, we have for the spring constant of the suspension:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$k = 4\pi^2 f^2 m = 4\pi^2(1.80\text{ s}^{-1})^2[1\,130\text{ kg} + 4(72.4\text{ kg})]$$

$$= 1.82 \times 10^5\text{ kg/s}^2$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

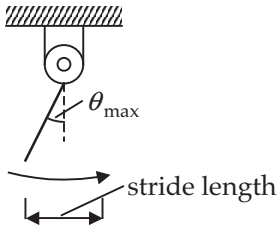
Now as the occupants exit,

$$x = \frac{F}{k} = \frac{4(72.4\text{ kg})(9.8\text{ m/s}^2)}{1.82\times10^5\text{ kg/s}^2} = \boxed{1.56\times10^{-2}\text{ m}}$$

***P15.62** (a) The period of the swinging rod is

$$T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{(1/3)m\ell^2}{mg\ell/2}}$$

$$= 2\pi\sqrt{\frac{2\ell}{3g}}$$



ANS. FIG. P15.62

The time for one half a cycle is $\frac{T}{2} = \pi\sqrt{\frac{2\ell}{3g}}$.

The distance traveled in this time is the stride length $2\ell \sin\theta_{max}$, so the speed is

$$\frac{d}{t} = \frac{2\ell\sin\theta_{max}}{\pi\sqrt{2\ell/3g}} = \frac{\sqrt{2\ell 3g}\sin\theta_{max}}{\pi} = \frac{\sqrt{6g\ell}\sin\theta_{max}}{\pi}$$

(b) We use the more precise expression

$$\frac{\sqrt{6g\ell\cos(\theta_{max}/2)}\sin\theta_{max}}{\pi}$$

$$= \frac{\sqrt{6(9.80\text{ m/s}^2)(0.850\text{ m})\cos 14.0^\circ}\sin 28.0^\circ}{\pi}$$

$$= \boxed{1.04\text{ m/s}}$$

(c) With

$$v_{old} = \frac{\sqrt{6g\ell_{old}\cos(\theta_{max}/2)}\sin\theta_{max}}{\pi}$$

$$v_{new} = \frac{\sqrt{6g\ell_{new}\cos(\theta_{max}/2)}\sin\theta_{max}}{\pi}$$

dividing gives

$$\frac{v_{new}}{v_{old}} = \frac{\sqrt{\ell_{new}}}{\sqrt{\ell_{old}}} = 2$$

$$\frac{\ell_{new}}{0.850\text{ m}} = 2^2$$

$$\ell_{new} = \boxed{3.40\text{ m}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

***P15.63** From $T = 2\pi\sqrt{\frac{L}{g}}$, the length of a pendulum with period T is $L = \frac{gT^2}{4\pi^2}$.

(a) On Earth, with T = 1.0 s,

$$L = \frac{gT^2}{4\pi^2} = \frac{(9.8\ \text{m/s}^2)(1.0\ \text{s})^2}{4\pi^2} = 0.25\ \text{m} = \boxed{25\ \text{cm}}$$

(b) If T = 1.0 s on Mars, then

$$L = \frac{g_{\text{Mars}}T^2}{4\pi^2} = \frac{(3.7\ \text{m/s}^2)(1.0\ \text{s})^2}{4\pi^2} = 0.094\ \text{m} = \boxed{9.4\ \text{cm}}$$

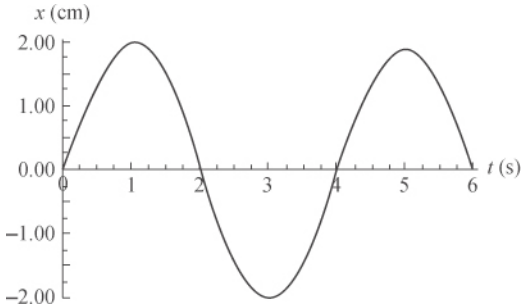
(c) and (d) The period of an object on a spring is $T = 2\pi\sqrt{\frac{m}{k}}$, which is independent of the local free-fall acceleration. Thus, the same mass will work on Earth and on Mars. This mass is

$$m = \frac{kT^2}{4\pi^2} = \frac{(10\ \text{N/m})(1.0\ \text{s})^2}{4\pi^2} = \boxed{0.25\ \text{kg}}$$

P15.64 (a) The amplitude is the magnitude of the maximum displacement from equilibrium (at x = 0). Thus, $\boxed{A = 2.00\ \text{cm}}$.

(b) The period is the time for one full cycle of the motion. Therefore, $\boxed{T = 4.00\ \text{s}}$.

(c) The angular frequency is $\omega = \frac{2\pi}{T} = \frac{2\pi}{4.00\ \text{s}} = \boxed{\frac{\pi}{2}\ \text{rad/s}}$.



ANS. FIG. P15.64

(d) The maximum speed is

$$v_{\max} = \omega A = \left(\frac{\pi}{2}\ \text{rad/s}\right)(2.00\ \text{cm}) = \boxed{\pi\ \text{cm/s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

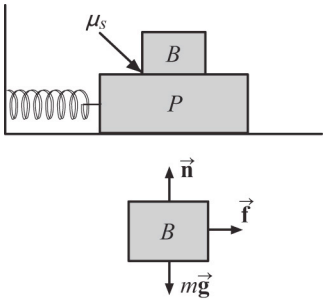
(e) The maximum acceleration is

$$a_{\max} = \omega^2 A = \left(\frac{\pi}{2}\ \text{rad/s}\right)^2 (2.00\ \text{cm}) = \boxed{4.93\ \text{cm/s}^2}$$

(f) The general equation for position as a function of time for an object undergoing simple harmonic motion with $x = 0$ when $t = 0$ and x increasing positively is $x = A\sin\omega t$. For this oscillator, this becomes

$$\boxed{x = 2.00\sin\left(\frac{\pi}{2}t\right), \text{ where } x \text{ is in centimeters and } t \text{ in seconds.}}$$

P15.65 The maximum acceleration of the oscillating system is $a_{\max} = A\omega^2 = 4\pi^2 Af^2$. The friction force exerted between the two blocks must be capable of accelerating Block B at this rate. Thus, if Block B is about to slip,



ANS. FIG. P15.65

$$f = f_{\max} = \mu_s n = \mu_s mg$$
$$= m\left(4\pi^2 Af^2\right)$$

which gives a maximum amplitude of oscillation of

$$A = \frac{\mu_s g}{4\pi^2 f^2} = \frac{(0.600)(980\ \text{cm/s}^2)}{4\pi^2(1.50\ \text{s}^{-1})^2} = \boxed{6.62\ \text{cm}}$$

P15.66 Refer to ANS. FIG. P15.65. The maximum acceleration of the oscillating system is $a_{\max} = A\omega^2 = 4\pi^2 Af^2$. The friction force exerted between the two blocks must be capable of accelerating Block B at this rate. Thus, if Block B is about to slip,

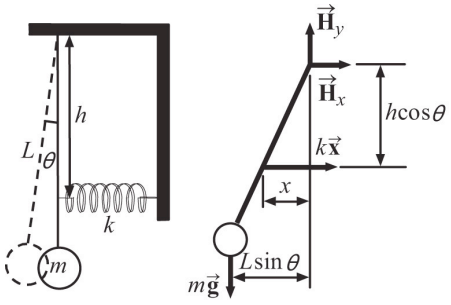
$$f = f_{\max} = \mu_s n = \mu_s mg = m\left(4\pi^2 Af^2\right)$$

which gives a maximum amplitude of oscillation of

$$A = \boxed{\frac{\mu_s g}{4\pi^2 f^2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.67 We draw a free-body diagram of the pendulum in ANS. FIG. P15.67. The force $\vec{\mathbf{H}}$ exerted by the hinge causes no torque about the axis of rotation.



ANS. FIG. P15.67

$$\tau = I\alpha \quad \text{and} \quad \frac{d^2\theta}{dt^2} = -\alpha$$

$$\tau = MgL\sin\theta + kxh\cos\theta$$

$$= -I\frac{d^2\theta}{dt^2}$$

For small-amplitude vibrations, use the approximations: $\sin\theta \approx \theta$, $\cos\theta \approx 1$, and $x \approx s = h\theta$.

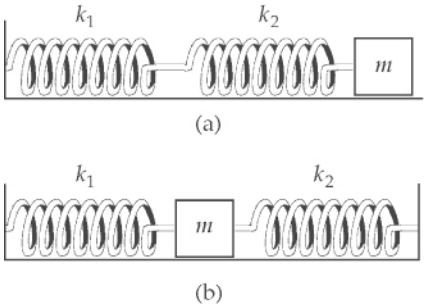
Therefore,

$$\frac{d^2\theta}{dt^2} = -\left(\frac{MgL + kh^2}{I}\right)\theta = -\omega^2\theta$$

$$\omega = \sqrt{\frac{MgL + kh^2}{ML^2}} = 2\pi f$$

$$f = \boxed{\frac{1}{2\pi L}\sqrt{gL + \frac{kh^2}{M}}}$$

P15.68 (a) When the mass is displaced a distance x from equilibrium, spring 1 is stretched a distance x_1 and spring 2 is stretched a distance x_2.



ANS. FIG. P15.68

By Newton's third law, we expect

$$k_1x_1 = k_2x_2$$

When this is combined with the requirement that

$$x = x_1 + x_2$$

we find $x_1 = \left[\frac{k_2}{k_1 + k_2}\right]x.$

The force on either spring is given by $F_1 = \left[\frac{k_1k_2}{k_1 + k_2}\right]x = ma$

where a is the acceleration of the mass m.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

This is in the form $F = k_{\text{eff}}x = ma$

and $T = 2\pi\sqrt{\dfrac{m}{k_{\text{eff}}}} = \boxed{2\pi\sqrt{\dfrac{m(k_1 + k_2)}{k_1 k_2}}}$.

(b) In this case each spring is distorted by the distance x which the mass is displaced. Therefore, the restoring force is

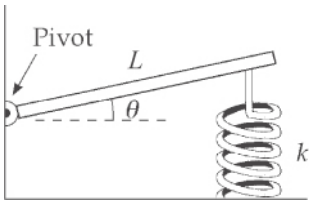
$$F = -(k_1 + k_2)x \quad \text{and} \quad k_{eff} = k_1 + k_2$$

so that $T = \boxed{2\pi\sqrt{\dfrac{m}{k_1 + k_2}}}$.

P15.69 At equilibrium, we have

$$\sum \tau = 0 - mg\left(\frac{L}{2}\right) + kx_0 L$$

where x_0 is the equilibrium compression.



ANS. FIG. P15.69

After displacement by a small angle (we assume $\cos\theta \approx 1$),

$$\sum \tau = -mg\left(\frac{L}{2}\right) + kxL = -mg\left(\frac{L}{2}\right) + k(x_0 - L\theta)L = -k\theta L^2$$

But,

$$\sum \tau = I\alpha = \frac{1}{3}mL^2\frac{d^2\theta}{dt^2}$$

so $$\frac{d^2\theta}{dt^2} = -\frac{3k}{m}\theta$$

Comparing this result to the general form for simple harmonic motion in which the angular acceleration is opposite in direction and proportional to the displacement,

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta$$

we find that

$$\omega^2 = \frac{3k}{m} \rightarrow \omega = \sqrt{\frac{3k}{m}} = \sqrt{\frac{3(100\ \text{N/m})}{5.00\ \text{kg}}} = \boxed{7.75\ \text{s}^{-1}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.70 Please refer to ANS. FIG. P15.69. At equilibrium, we have

$$\sum \tau = 0 - mg\left(\frac{L}{2}\right) + kx_0 L$$

where x_0 is the equilibrium compression.

After displacement by a small angle (we assume $\cos\theta \approx 1$),

$$\sum \tau = -mg\left(\frac{L}{2}\right) + kxL = -mg\left(\frac{L}{2}\right) + k\left(x_0 - L\theta\right)L = -k\theta L^2$$

But,

$$\sum \tau = I\alpha = \frac{1}{3}mL^2\frac{d^2\theta}{dt^2}$$

Comparing this result to the general form for simple harmonic motion in which the angular acceleration is opposite in direction and proportional to the displacement,

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta$$

we find that

$$\omega^2 = \frac{3k}{m} \rightarrow \boxed{\omega = \sqrt{\frac{3k}{m}}}$$

P15.71 As it passes through equilibrium, the 4.00-kg object has speed

$$v_{\max} = \omega A = \sqrt{\frac{k}{m}}A = \sqrt{\frac{100\ \text{N/m}}{4.00\ \text{kg}}}(2.00\ \text{m}) = 10.0\ \text{m/s}$$

In the completely inelastic collision, momentum of the two-object system is conserved. So the new 10.0-kg object starts its oscillation with a new maximum speed given by

$$(4.00\ \text{kg})(10.0\ \text{m/s}) + (6.00\ \text{kg})0 = (10.0\ \text{kg})v_{\max}$$
$$v_{\max} = 4.00\ \text{m/s}$$

(a) The system consisting of the two objects, the spring, and the Earth, is isolated, so mechanical energy is conserved. The new amplitude is given by

$$\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2$$

$$(10.0\ \text{kg})(4.00\ \text{m/s})^2 = (100\ \text{N/m})A^2$$

$$A = \boxed{1.26\ \text{m}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) The old period was $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{4.00\text{ kg}}{100\text{ N/m}}} = 1.26\text{ s}.$

The new period is $T = 2\pi\sqrt{\frac{10}{100}\text{ s}^2} = 1.99\text{ s}.$

The period of the system has changed by a factor of

$$\frac{f_{\text{new}}}{f_{\text{old}}} = \frac{1.99\text{ s}}{1.26\text{ s}} = \boxed{1.58}$$

(c) The old energy was $\frac{1}{2}mv_{\max}^2 = \frac{1}{2}(4.00\text{ kg})(10.0\text{ m/s})^2 = 200\text{ J}.$

The new mechanical energy is $\frac{1}{2}(10.0\text{ kg})(4.00\text{ m/s})^2 = 80.0\text{ J}.$

The energy has $\boxed{\text{decreased by 120 J}}$.

(d) $\boxed{\text{Mechanical energy is transformed into internal energy in the perfectly inelastic collision.}}$

P15.72 (a) $\sum\vec{\mathbf{F}} = -2T\sin\theta\,\hat{\mathbf{j}}$

where $\theta = \tan^{-1}\left(\frac{y}{L}\right)$.



ANS. FIG. P15.72

Therefore, for a small displacement,

$\sin\theta \approx \tan\theta = \frac{y}{L}$ and $\boxed{\sum\vec{\mathbf{F}} = \frac{-2Ty}{L}\hat{\mathbf{j}}}$

(b) The total force exerted on the ball is opposite in direction and proportional to its displacement from equilibrium, so the ball moves with simple harmonic motion. For a spring system,

$$\sum\vec{\mathbf{F}} = -k\vec{\mathbf{x}} \quad \text{becomes here} \quad \sum\vec{\mathbf{F}} = -\frac{2T}{L}\vec{\mathbf{y}}.$$

Therefore, the effective spring constant is $\frac{2T}{L}$ and

$$\boxed{\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2T}{mL}}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.73 One can write the following equations of motion:

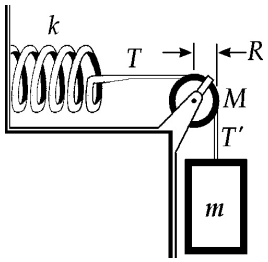
$$T - kx = 0 \qquad \text{(describes the spring)}$$

$$mg - T' = ma = m\frac{d^2x}{dt^2}$$

(for the hanging object)

$$R(T' - T) = I\frac{d^2\theta}{dt^2} = \frac{I}{R}\frac{d^2x}{dt^2}$$

(for the pulley)



ANS. FIG. P15.73

with $I = \frac{1}{2}MR^2$.

Combining these equations gives the equation of motion:

$$\left(m + \frac{1}{2}M\right)\frac{d^2x}{dt^2} + kx = mg$$

The solution is $x(t) = A\sin\omega t + \frac{mg}{k}$ (where $\frac{mg}{k}$ arises because of the extension of the spring due to the weight of the hanging object), with angular frequency

$$\omega = \sqrt{\frac{k}{m + \frac{1}{2}M}} = \sqrt{\frac{2k}{2m + M}}$$

(a) For k = 100 N/m and m = 0.200 kg,

$$\boxed{\omega = \sqrt{\frac{200}{0.400 + M}}, \text{ where } \omega \text{ is in s}^{-1} \text{ and } M \text{ is in kilograms.}}$$

(b) The highest possible value occurs when M = 0: $\omega = \boxed{22.4\ \text{s}^{-1}}$.

(c) The angular frequency is independent of the radius of the pulley: $\omega = \boxed{22.4\ \text{s}^{-1}}$

P15.74 Suppose a 100-kg biker compresses the suspension 2.00 cm.

Then,

$$k = \frac{F}{x} = \frac{980\ \text{N}}{2.00 \times 10^{-2}\ \text{m}} = 4.90 \times 10^4\ \text{N/m}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

If total mass of motorcycle and biker is 500 kg, the frequency of free vibration is

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{4.90\times10^4\ \text{N/m}}{500\ \text{kg}}} = 1.58\ \text{Hz}$$

If he encounters washboard bumps at the same frequency as the free vibration, resonance will make the motorcycle bounce a lot. It may bounce so much as to interfere with the rider's control of the machine.

Assuming a speed of 20.0 m/s, we find these ridges are separated by

$$\frac{20.0\ \text{m/s}}{1.58\ \text{s}^{-1}} = 12.7\ \text{m}\ \boxed{\sim 10^1\ \text{m}}$$

In addition to this vibration mode of bouncing up and down as one unit, the motorcycle can also vibrate at higher frequencies by rocking back and forth between front and rear wheels, by having just the front wheel bounce inside its fork, or by doing other things. Other spacing of bumps will excite all of these other resonances.

P15.75 (a) $T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{L}{g}} = 2\pi\sqrt{\dfrac{2.23\ \text{m}}{9.80\ \text{m/s}^2}} = \boxed{3.00\ \text{s}}$

(b) $E = \dfrac{1}{2}mv^2 = \dfrac{1}{2}(6.74\ \text{kg})(2.06\ \text{m/s})^2 = \boxed{14.3\ \text{J}}$

(c) For a system of an isolated pendulum-Earth, mechanical energy is conserved. Relate the pendulum bob at the lowest point to the highest point:

$$\Delta K + \Delta U_g = 0$$

$$\left(0 - \frac{1}{2}mv^2\right) + (mgh - 0) = 0$$

$$mgh = \frac{1}{2}mv^2$$

$$h = \frac{v^2}{2g} = \frac{(2.06\ \text{m/s})^2}{2(9.80\ \text{m/s}^2)} = 0.217\ \text{m}$$

and

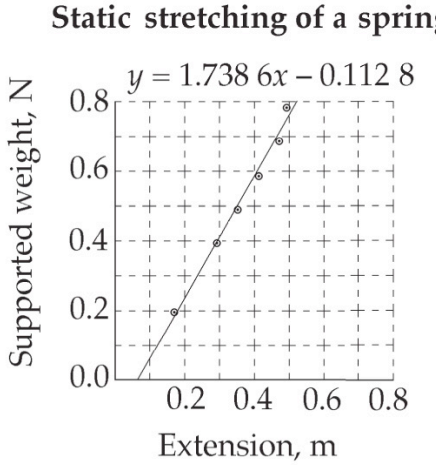
$$h = L - L\cos\theta = L(1-\cos\theta)$$

$$\cos\theta = 1 - \frac{h}{L} = 1 - \frac{0.217\ \text{m}}{2.23\ \text{m}}$$

$$\boxed{\theta = 25.5^\circ}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.76 (a) The graph of Mg versus x is shown in ANS. FIG. P15.76(a).



ANS. FIG. P15.76(a)

(b) Assuming a Hooke's Law type spring, $F = Mg = kx$, and empirically

$$Mg = 1.74x - 0.113$$

so $k = \boxed{1.74 \text{ N/m} \pm 6\%}$

(c)

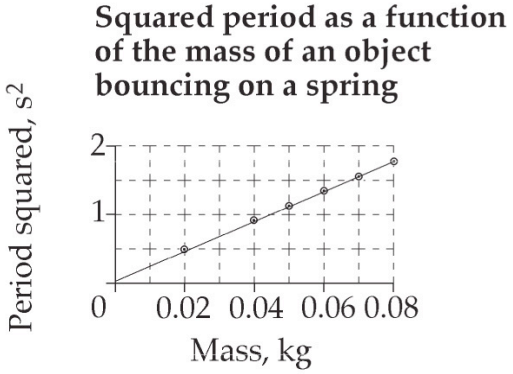
M, kg	x, m	Mg, N
0.020 0	0.17	0.196
0.040 0	0.293	0.392
0.050 0	0.353	0.49
0.060 0	0.413	0.588
0.070 0	0.471	0.686
0.080 0	0.493	0.784

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d)

Time, s	T, s	M, kg	T^2, s^2
7.03	0.703	0.020 0	0.494
9.62	0.962	0.040 0	0.925
10.67	1.067	0.050 0	1.138
11.67	1.167	0.060 0	1.362
12.52	1.252	0.070 0	1.568
13.41	1.341	0.080 0	1.798

(e) The graph of T^2 versus M is shown in ANS. FIG. P15.76(e).



ANS. FIG. P15.76(e)

(f) We may write the equation as theoretically

$$T^2 = \frac{4\pi^2}{k} M + \frac{4\pi^2}{3k} m_s$$

and empirically

$$T^2 = 21.7\, M + 0.058\, 9$$

so

$$k = \frac{4\pi^2}{21.7} = \boxed{1.82 \text{ N/m} \pm 3\%}$$

(g) The k values $1.74 \text{ N/m} \pm 6\%$

and $1.82 \text{ N/m} \pm 3\%$ differ by 4% so they agree.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(h) Utilizing the axis-crossing point,

$$m_s = 3\left(\frac{0.0589}{21.7}\right)\text{ kg} = \boxed{8\text{ grams} \pm 12\%}$$

$\boxed{\text{in agreement}}$ with 7.4 grams.

P15.77 The free-body diagram in ANS. FIG. P15.77 shows the forces acting on the balloon when it is displaced distance $s = L\theta$ along the circular arc it follows. The net force tangential to this path is

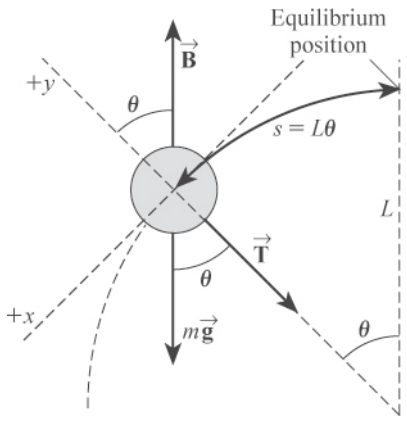
$$F_{\text{net}} = \sum F_x = -B\sin\theta + mg\sin\theta = -\left(B - mg\right)\sin\theta$$

For small angles, $\sin\theta \approx \theta = s / L$

Also, $mg = \left(\rho_{\text{He}} V\right) g$

and the buoyant force is $B = \left(\rho_{\text{air}} V\right) g$. Thus, the net restoring force acting on the balloon is

$$F_{\text{net}} \approx -\left[\frac{\left(\rho_{\text{air}} - \rho_{\text{He}}\right) Vg}{L}\right] s$$



ANS. FIG. P15.77

Observe that this is in the form of Hooke's law, $F = -k\,s$, with

$$k = \left(\rho_{\text{air}} - \rho_{\text{He}}\right) Vg / L$$

Thus, the motion will be simple harmonic and the period is given by

$$T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{\rho_{\text{He}} V}{\left(\rho_{\text{air}} - \rho_{\text{He}}\right) Vg / L}}$$

$$= 2\pi\sqrt{\left(\frac{\rho_{\text{He}}}{\rho_{\text{air}} - \rho_{\text{He}}}\right)\frac{L}{g}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

This yields

$$T = 2\pi\sqrt{\left(\frac{0.179\ \text{kg/m}^3}{1.20\ \text{kg/m}^3 - 0.179\ \text{kg/m}^3}\right)\frac{(3.00\ \text{m})}{(9.80\ \text{m/s}^2)}} = \boxed{1.46\ \text{s}}$$

P15.78 (a) We require $Ae^{-bt/2m} = \dfrac{A}{2} \quad \rightarrow \quad e^{+bt/2m} = 2,$

or $\dfrac{bt}{2m} = \ln 2$

or $\dfrac{0.100\ \text{kg/s}}{2(0.375\ \text{kg})}t = 0.693$

which gives $t = \boxed{5.20\ \text{s}}$.

The spring constant is irrelevant.

(b) We can evaluate the energy at successive turning points, where

$\cos(\omega t + \phi) = \pm 1$ and the energy is $\dfrac{1}{2}kx^2 = \dfrac{1}{2}kA^2e^{-bt/m}$.

We require $\dfrac{1}{2}kA^2e^{-bt/m} = \dfrac{1}{2}\left(\dfrac{1}{2}kA^2\right)$

or $e^{+bt/m} = 2$

which gives

$$t = \frac{m(\ln 2)}{b} = \frac{(0.375\ \text{kg})(0.693)}{0.100\ \text{kg/s}} = \boxed{2.60\ \text{s}}$$

(c) From $E = \dfrac{1}{2}kA^2$, the fractional rate of change of energy over time is

$$\frac{dE/dt}{E} = \frac{(d/dt)\left(\frac{1}{2}kA^2\right)}{\frac{1}{2}kA^2} = \frac{\frac{1}{2}k(2A)(dA/dt)}{\frac{1}{2}kA^2} = 2\frac{dA/dt}{A}$$

which gives $\boxed{\dfrac{dA/dt}{A} = \dfrac{1}{2}\dfrac{dE/dt}{E}}$.

which is twice as fast as the fractional rate of change in amplitude.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.79 (a) $x = A\cos(\omega t + \phi) \rightarrow v = -\omega A \sin(\omega t + \phi)$

We have at, $t = 0,\ v = -\omega A \sin\phi = -v_{\max}$.

This requires $\phi = 90°,$ so $x = A\cos(\omega t + 90°)$

$\rightarrow x = A\cos\left(\omega t + \frac{\pi}{2}\right).$

Numerically we have $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{50.0\ \text{N/m}}{0.500\ \text{kg}}} = 10.0\ \text{s}^{-1}$

and $v_{\max} = \omega A \rightarrow 20.0\ \text{m/s} = (10.0\ \text{s}^{-1})A \rightarrow A = 2.00\ \text{m}.$

So $\boxed{x = 2\cos\left(10t + \frac{\pi}{2}\right)}$, where x is in meters and t in seconds.

(b) Using $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2,$ we require $\frac{1}{2}kx^2 = 3\left(\frac{1}{2}mv^2\right)$

which implies $\frac{1}{3}\left(\frac{1}{2}kx^2\right) + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \rightarrow \frac{4}{3}x^2 = A^2$

which gives $x = \pm\sqrt{\frac{3}{4}}A = \pm 0.866(2.00\ \text{m}) = \boxed{\pm 1.73\ \text{m}}$

(c) The particle's position is given by $x = 2\cos\left(10t + \frac{\pi}{2}\right).$

The particle is at $x = 0$ when

$$10t + \frac{\pi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots \rightarrow\ 10t = 0,\ \pi,\ 2\pi, 4\pi \ldots$$

At $t = 0$, the particle is at the origin, but moving to the left. The next time the particle is at the origin is when $10t = \pi$ when it is moving to the right.

The particle is first at $x = 1.00$ m when $10t + \frac{\pi}{2} = \frac{3\pi}{2} + \frac{\pi}{3} = \frac{11\pi}{6}.$

So then, $10t = \frac{4\pi}{3}.$

The minimum time required for the particle to move from $x = 0$ to $x = 1.00$ m is

$$10\Delta t = \frac{4\pi}{3} - \pi = \frac{\pi}{3} \rightarrow \Delta t = \frac{\pi}{30} = \boxed{0.105\ \text{s} = 105\ \text{ms}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d) $$\omega = \sqrt{\frac{g}{L}} \rightarrow L = \frac{g}{\omega^2} = \frac{9.80 \text{ m/s}^2}{\left(10 \text{ s}^{-1}\right)^2} = \boxed{0.098\ 0 \text{ m}}$$

P15.80 (a) The block moves with the board in what we take as the positive x direction, stretching the spring until the spring force $-kx$ is equal in magnitude to the maximum force of static friction:

$$kx = \mu_s n = \mu_s mg$$

This occurs at $x = \dfrac{\mu_s mg}{k}$.

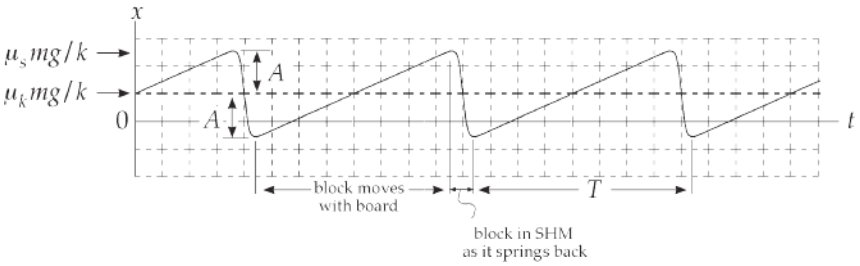
(b) Since v is small, the block is nearly at the rest at this break point. It starts almost immediately to move back to the left, the forces on it being $-kx$ and $+\mu_k mg$. While it is sliding the net force exerted on it can be written as

$$F_{\text{net}} = -kx + \mu_k mg = -kx + \frac{k\mu_k mg}{k} = -k\left(x - \frac{\mu_k mg}{k}\right)$$
$$= -kx_{rel}$$

where x_{rel} is the excursion of the block away from the point $\dfrac{\mu_k mg}{k}$.

Conclusion: the block goes into simple harmonic motion centered about the equilibrium position where the spring is stretched by $\dfrac{\mu_k mg}{k}$.

(c) The graph of the motion looks as shown in ANS. FIG. P15.80(c):



ANS. FIG. P15.80(c)

(d) The amplitude of its motion is its original displacement, $A = \dfrac{\mu_s mg}{k} - \dfrac{\mu_k mg}{k}$, because the block has been pulled out to $x = \dfrac{\mu_s mg}{k}$, then it goes into simple harmonic motion centered about $x = \dfrac{\mu_k mg}{k}$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

It first comes to rest at spring extension $\frac{\mu_k mg}{k} - A = \frac{(2\mu_k - \mu_s)mg}{k}$.

Almost immediately at this point it latches onto the slowly-moving board to move with the board. The board exerts a force of static friction on the block, and the cycle continues.

(e) The time during each cycle when the block is moving with the board is $\frac{2A}{v} = \frac{2(\mu_s - \mu_k)mg}{kv}$. The time for which the block is springing back is one half a cycle of simple harmonic motion, $\frac{1}{2}\left(2\pi\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{m}{k}}\right)$ (because the block slides from $+A$ to $-A$ during its SHM). We ignore the times at the end points of the motion when the speed of the block changes from v to 0 and from 0 to v. Since v is small compared to $\frac{2A}{\pi\sqrt{m/k}}$, these times are negligible. Then the period is

$$T = \frac{2(\mu_s - \mu_k)mg}{kv} + \pi\sqrt{\frac{m}{k}}$$

P15.81 (a) Let ℓ represent the length below water at equilibrium and M the tube's mass:

$$\sum F_y = 0 \Rightarrow -Mg + \rho\pi r^2 \ell g = 0$$

Now with any excursion x from equilibrium

$$-Mg + \rho\pi r^2 (\ell - x) g = Ma$$

Subtracting the equilibrium equation gives

$$-\rho\pi r^2 gx = Ma \rightarrow a = -\left(\frac{\rho\pi r^2 g}{M}\right)x$$

The opposite direction and direct proportionality of a to x imply SHM.

(b) For SHM, $F = -kx = ma \rightarrow a = -(k/m)x = -\omega^2 x$: the coefficient of x is the square of the angular frequency:

$$\omega = \sqrt{\frac{\rho\pi r^2 g}{M}} \rightarrow T = \frac{2\pi}{\omega} = \boxed{\frac{2}{r}\sqrt{\frac{\pi M}{\rho g}}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.82 From the oscillator information, find the natural frequency of the oscillator:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{10.0\ \text{N/m}}{0.001\ \text{kg}}} = 100\ \text{s}^{-1}$$

From the measurement information, find the value of $b/2m$:

$$\frac{x_{\max}(23.1\ \text{ms})}{x_{\max}(0)} = 0.250 = \frac{Ae^{-(b/2m)(0.023\ 1\ \text{s})}}{A\left(e^0\right)} = e^{-(b/2m)(0.023\ 1\ \text{s})}$$

Solving,

$$\frac{b}{2m} = -\frac{\ln(0.250)}{0.0231\ \text{s}} = 60.0\ \text{s}^{-1}$$

If the damping constant is doubled, $b/2m = 120\ \text{s}^{-1}$. In this case, however, $b/2m > \omega_0$ and the system is overdamped. Your design objective is not met because the system does not oscillate.

P15.83 The effective spring constant of a ball is

$$k = \frac{|F|}{|x|} = \frac{1.60\times10^{3}\ \text{N}}{0.200\times10^{-3}\ \text{m}} = 80.0\ \text{MN/m}$$

The half-cycle is from the equilibrium position of the model spring to maximum compression and back to equilibrium again. The time is one-half the period:

$$\frac{1}{2}T = \frac{1}{2}(2\pi)\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{0.067\ 4\ \text{kg}}{80.0\times10^{6}\ \text{N/m}}} = \boxed{9.12\times10^{-5}\ \text{s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Challenge Problems

P15.84 (a) $\Delta K + \Delta U = 0$

Thus, $K_{\text{top}} + U_{\text{top}} = K_{\text{bot}} + U_{\text{bot}}$

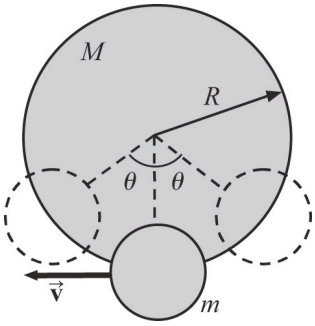
where $K_{\text{top}} = U_{\text{bot}} = 0.$

Therefore, $mgh = \frac{1}{2} I\omega^2$, but

$$h = R - R\cos\theta = R(1-\cos\theta)$$

$$\omega = \frac{v}{R}$$

and $I = \frac{MR^2}{2} + \frac{mr^2}{2} + mR^2$



ANS. FIG. P15.84

Substituting, we find

$$mgR(1-\cos\theta) = \frac{1}{2}\left(\frac{MR^2}{2} + \frac{mr^2}{2} + mR^2\right)\frac{v^2}{R^2}$$

$$mgR(1-\cos\theta) = \left[\frac{M}{4} + \frac{mr^2}{4R^2} + \frac{m}{2}\right]v^2$$

and $v^2 = 4gR\left(\frac{1-\cos\theta}{M/m + r^2/R^2 + 2}\right)$, so

$$\boxed{v = 2\left[\frac{Rg(1-\cos\theta)}{M/m + r^2/R^2 + 2}\right]^{1/2}}$$

(b) $T = 2\pi\sqrt{\frac{I}{m_T g d_{\text{CM}}}}$

Substituting $m_T = m + M$ and solving for d_{CM} gives

$$d_{\text{CM}} = \frac{mR + M(0)}{m + M}$$

The period is then

$$T = 2\pi\sqrt{\frac{\frac{1}{2}MR^2 + \frac{1}{2}mr^2 + mR^2}{mgR}} = 2\pi\sqrt{\frac{\frac{1}{2}\left(MR^2 + 2mR^2 + mr^2\right)}{mgR}}$$

$$= \boxed{2\pi\left[\frac{(M+2m)R^2 + mr^2}{2mgR}\right]^{1/2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.85 (a) Total energy $= \frac{1}{2}kA^2 = \frac{1}{2}(100\ \text{N/m})(0.200\ \text{m})^2 = 2.00\ \text{J}.$

At equilibrium, the total energy is:

$$\frac{1}{2}(m_1 + m_2)v^2 = \frac{1}{2}(16.0\ \text{kg})v^2 = (8.00\ \text{kg})v^2$$

Therefore,

$$(8.00\ \text{kg})v^2 = 2.00\ \text{J}, \text{ and } v = \boxed{0.500\ \text{m/s}}$$

This is the speed of m_1 and m_2 at the equilibrium point. Beyond this point, the mass m_2 moves with the constant speed of 0.500 m/s while mass m_1 starts to slow down due to the restoring force of the spring.

(b) The energy of the m_1-spring system at equilibrium is:

$$\frac{1}{2}m_1v^2 = \frac{1}{2}(9.00\ \text{kg})(0.500\ \text{m/s})^2 = 1.125\ \text{J}$$

This is also equal to $\frac{1}{2}k(A')^2$, where A' is the amplitude of the m_1-spring system.

Therefore,

$$\frac{1}{2}(100)(A')^2 = 1.125 \text{ or } A' = 0.150\ \text{m}$$

The period of the m_1-spring system is

$$T = 2\pi\sqrt{\frac{m_1}{k}} = 2\pi\sqrt{\frac{9.00\ \text{kg}}{100\ \text{N/m}}} = 1.885\ \text{s}$$

and it takes $\frac{1}{4}T = 0.471\ \text{s}$ after it passes the equilibrium point for the spring to become fully stretched the first time. The distance separating m_1 and m_2 at this time is

$$D = v\left(\frac{T}{4}\right) - A' = (0.500\ \text{m/s})(0.471\ \text{s}) - 0.150\ \text{m}$$

$$= 0.085\ 6\ \text{m} = \boxed{8.56\ \text{cm}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.86 The time interval for your competitor's package to arrive is half of the orbital period found from Kepler's third law, Equation 13.11:

$$\Delta t = \frac{1}{2}T = \frac{1}{2}\sqrt{\frac{4\pi^2}{GM_E}\left(R_E\right)^3} = \pi\sqrt{\frac{R_E^3}{GM_E}}$$

Now, consider your proposal. The force on the package at an arbitrary position r is

$$F_g = -G\frac{M_{\text{closer than } r}m}{r^2} = -G\frac{m}{r^2}\frac{\left(\frac{4}{3}\pi r^3\right)}{\left(\frac{4}{3}\pi R_E^3\right)}M_E = -G\frac{M_E m}{R_E^3}r$$

This force is of the form of Hooke's law! The "spring constant" for this motion is

$$k = G\frac{M_E m}{R_E{}^3}$$

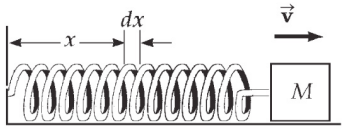
Because the force on the package is a Hooke's-law force, the package will oscillate between opposite points on the Earth in simple harmonic motion. To deliver the package to the other side of the Earth, someone must grab the package before it begins its return journey. The time interval for the package to travel to the other side of the Earth is half of a period of oscillation:

$$\Delta t = \frac{1}{2}T = \frac{1}{2}\left[2\pi\sqrt{\frac{m}{k}}\right] = \frac{1}{2}\left[2\pi\sqrt{m\left(G\frac{M_E m}{R_E^3}\right)^{-1}}\right] = \pi\sqrt{\frac{R_E^3}{GM_E}}$$

This is exactly the same time interval as for your competitor, so you have no advantage! In fact, you have the disadvantage of the initial capital outlay to bore through the entire Earth!

P15.87 (a) For each segment of the spring:

$$dK = \frac{1}{2}(dm)v_x^2$$



ANS. FIG. P15.87

Also,

$$v_x = \frac{x}{\ell}v \quad \text{and} \quad dm = \frac{m}{\ell}dx$$

Therefore, the total kinetic energy of the block-spring system is

$$K = \frac{1}{2}Mv^2 + \frac{1}{2}\int_0^{\ell}\left(\frac{x^2v^2}{\ell^2}\right)\frac{m}{\ell}dx = \boxed{\frac{1}{2}\left(M + \frac{m}{3}\right)v^2}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) $\omega = \sqrt{\dfrac{k}{m_{\text{eff}}}}$ and $\dfrac{1}{2}m_{\text{eff}}v^2 = \dfrac{1}{2}\left(M + \dfrac{m}{3}\right)v^2$

Therefore,

$$T = \frac{2\pi}{\omega} = \boxed{2\pi\sqrt{\frac{M + m/3}{k}}}$$

P15.88 (a) Note that as the spring passes through the vertical position, the object is moving in a circular arc of radius $L - y_f$, where the y coordinate of the object at this point must be negative ($y_f < 0$). When the object is at y_f, the spring is stretched $x = y_f - L$. At position y_f, the spring is stretched and exerting an upward tension force of magnitude greater than the object's weight. This is necessary so the object experiences a net force toward the pivot to supply the needed centripetal acceleration in this position. This is summarized by Newton's second law applied to the object at this point, stating (remember, y_f is negative)

$$\sum F_y = ma \rightarrow -ky_f - mg = \frac{mv^2}{L - y_f} \qquad \textbf{[1]}$$

The system is isolated, so conservation of energy requires that

$$E = KE_i + PE_{g,i} + PE_{s,i} = KE_f + PE_{g,f} + PE_{s,f}$$

or

$$E = 0 + mgL + 0 = \frac{1}{2}mv^2 + mgy_f + \frac{1}{2}ky_f^2$$

reducing to

$$2mg\left(L - y_f\right) = mv^2 + ky_f^2 \qquad \textbf{[2]}$$

From equation [1], observe that $mv^2 = -(L - y_f)(ky_f + mg)$.

Substituting this into equation [2] gives

$$2mg(L - y_f) = -(L - y_f)(ky_f + mg) + ky_f$$

After expanding and regrouping terms, this becomes

$$(2k)y_f^2 + (3mg - kL)y_f + (-3mgL) = 0$$

which is a quadratic equation $ay_f^2 + by_f + c = 0$, with

$$a = 2k = 2\left(1250\ \text{N/m}\right) = 2.50 \times 10^3\ \text{N/m}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

$$b = 3mg - kL = 3(5.00\text{ kg})(9.80\text{ m/s}^2) - (1\,250\text{ N/m})(1.50\text{ m})$$
$$= -1.73\times10^3\text{ N}$$

and

$$c = -3mgL = -3(5.00\text{ kg})(9.80\text{ m/s}^2)(1.50\text{ m}) = -221\text{ N}\cdot\text{m}$$

Applying the quadratic formula, keeping only the negative solution [see the discussion in part (a)], and suppressing units, gives

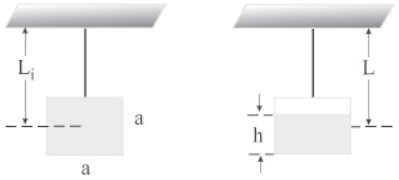
$$y_f = \frac{-b-\sqrt{b^2-4ac}}{2a}$$
$$= \frac{-\left(-1.73\times10^3\right)-\sqrt{\left(-1.73\times10^3\right)^2-4\left(2.50\times10^3\right)(-221)}}{2\left(2.50\times10^3\right)}$$

or $\boxed{y_f = -0.110\text{ m}}$

(b) Because the length of this pendulum varies and is longer throughout its motion than a simple pendulum of length L, $\boxed{\text{its period will be longer}}$ than that of a simple pendulum.

P15.89 (a) The period of the pendulum is given by

$$T = 2\pi\sqrt{\frac{L}{g}}$$



ANS. FIG. P15.89

and changes as

$$\frac{dT}{dt} = \frac{\pi}{\sqrt{g}}\frac{1}{\sqrt{L}}\frac{dL}{dt} \qquad \textbf{[1]}$$

We need to find $L(t)$ and $\frac{dL}{dt}$. From the diagram in ANS. FIG. P15.89(a),

$$L = L_i + \frac{a}{2} - \frac{h}{2} \qquad \text{and} \qquad \frac{dL}{dt} = -\left(\frac{1}{2}\right)\frac{dh}{dt}$$

But $\frac{dM}{dt} = \rho\frac{dV}{dt} = -\rho A\frac{dh}{dt}$. Therefore,

$$\frac{dh}{dt} = -\frac{1}{\rho A}\frac{dM}{dt} \qquad \rightarrow \qquad \frac{dL}{dt} = \left(\frac{1}{2\rho A}\right)\frac{dM}{dt} \qquad \textbf{[2]}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Also,

$$\int_{L_i}^{L} dL = \left(\frac{1}{2\rho A}\right)\left(\frac{dM}{dt}\right)t = L - L_i \qquad \textbf{[3]}$$

Substituting equations [2] and [3] into [1] gives:

$$\frac{dT}{dt} = \frac{\pi}{\sqrt{g}}\left(\frac{1}{2\rho a^2}\right)\left(\frac{dM}{dt}\right)\frac{1}{\sqrt{L_i + \left(t / 2\rho a^2\right)(dM/dt)}}$$

Integrating, we get

$$T = \frac{\pi}{\sqrt{g}}\left(\frac{1}{2\rho a^2}\right)\left(\frac{dM}{dt}\right)\int_0^t \frac{dt}{\sqrt{L_i + \left(t / 2\rho a^2\right)(dM/dt)}}$$

$$T = \frac{\pi}{\sqrt{g}}\left(\frac{1}{2\rho a^2}\right)\left(\frac{dM}{dt}\right)\frac{2\sqrt{L_i + \left(t / 2\rho a^2\right)(dM/dt)}}{\left(1/2\rho a^2\right)(dM/dt)}$$

$$T = \boxed{\frac{2\pi}{\sqrt{g}}\sqrt{L_i + \frac{1}{2\rho a^2}\left(\frac{dM}{dt}\right)t}}$$

(b) When the liquid is gone, the CM of the bob is suddenly again at the center of the cube. We had ignored the mass of the cube up until now since it was small compared to the mass of the liquid. Thus, once the liquid is gone, $L = L_i$.

$$T = \boxed{2\pi\sqrt{\frac{L_i}{g}}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

ANSWERS TO EVEN-NUMBERED PROBLEMS

P15.2 1.59 k N/m

P15.4 (a) 4.33 cm; (b) –5.00 cm/s; (c) –17.3 cm/s^2; (d) 3.14 s; (e) 5.00 cm

P15.6 (a) 18.8 m/s; (b) 7.11 km/s^2

P15.8 (a) 2.40 s; (b) 0.417 Hz; (c) 2.62 rad/s

P15.10 39.2 N

P15.12 (a) 15.8 cm; (b) 51.1 m; (c) –15.9 cm; (d) 50.8 m; (e) The patterns of oscillation diverge from each other, starting out in phase but becoming completely out of phase. To calculate the future, we would need *exact* knowledge of the present; an impossibility.

P15.14 (a) motion is periodic; (b) 1.81 s; (c) The motion is not simple harmonic. The net force acting on the ball is a constant given by $F = -mg$ (except when it is in contact with the ground), which is not in the form of Hooke's law.

P15.16 (a) See P15.16(a) for complete solution; (b) See P15.16(b) for complete solution

P15.18 (a) 1.26 s; (b) 0.150 m/s, 0.750 m/s^2; (c) $x = 3.00\cos(5.00t + \pi)$, $-15.0\sin(5.00t + \pi)$, and $-75.0\cos(5.00t + \pi)$

P15.20 (a) yes; (b) We see that finding the period does not depend on knowing the mass: $T = 0.859$ s.

P15.22 (a) 126 N/m; (b) 0.178 m

P15.24 (a) 0.153 J; (b) 0.784 m/s; (c) 17.5 m/s^2

P15.26 (a) E increases by a factor of 4; (b) v_{max} is doubled; (c) a_{max} also doubles; (d) the period is unchanged.

P15.28 (a) 100 N/m; (b) 1.13 Hz; (c) 1.41 m/s; (d) $x = 0$; (e) 10.0 m/s^2; (f) ±0.200 m; (g) 2.00 J; (h) 1.33 m/s; (i) 3.33 m/s^2

P15.30 (a) Particle under constant acceleration; (b) 1.50 s; (c) isolated; (d) 73.4 N/m; (e) 19.7 m below the bridge; (f) 1.06 rad/s; (g) +2.01 s; (h) 3.50 s

P15.32 (a) 5.98 m/s; (b) 206 N/m; (c) 0.238 m

P15.34 1.001 5

P15.36 $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{R}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.38 $I = \dfrac{mgd}{4\pi^2 f^2}$

P15.40 (a) $2\pi\sqrt{\dfrac{\left(I_{\text{CM}} + md^2\right)}{mgd}}$; (b) $I_{\text{CM}} = md^2$

P15.42 (a) 2.09 s; (b) 4.08%

P15.44 For Length, L (m): 1.000, 0.750, 0.500 and Period, T (s): 2.00, 1.73, 1.42; (b) For Period T(s): 2.00, 1.73, 1.42 and g (m/s^2): 9.87, 9.89, 9.79. This agrees with the accepted value of g = 9.80 m/s^2 within 0.5%; (c) 9.94 m/s^2

P15.46 $1.00 \times 10^{-3}\ \text{s}^{-1}$

P15.48 $\dfrac{dE}{dt} = -bv^2 < 0$

P15.50 (a) 1.19 Hz; (b) 17.5 cm

P15.52 318 N

P15.54 See P15.54 for complete solution.

P15.56 0.919×10^{14} Hz

P15.58 (a) 0.368 m/s; (b) 3.51 cm; (c) 40.6 mJ; (d) 27.7 mJ

P15.60 (a) 4.31 cm; (b) When the rock is on the point of lifting off, the surrounding water is also barely in free fall. No pressure gradient exists in the water, so no buoyant force acts on the rock. The effect of the surrounding water disappears at that instant.

P15.62 (a) See P15.62(a) for complete solution; (b) 1.04 m/s; (c) 3.40 m

P15.64 (a) A = 2.00 cm; (b) T = 4.00 s; (c) $\dfrac{\pi}{2}$ rad/s; (d) π cm/s; (e) 4.93 cm/s^2; (f) $x = 2.00\sin\left(\dfrac{\pi}{2}t\right)$, where x is in centimeters and t is in seconds

P15.66 $\dfrac{\mu_s g}{4\pi^2 f^2}$

P15.68 (a) $2\pi\sqrt{\dfrac{m\left(k_1 + k_2\right)}{k_1 k_2}}$; (b) $2\pi\sqrt{\dfrac{m}{\left(k_1 + k_2\right)}}$

P15.70 $\omega = \sqrt{\dfrac{3k}{m}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P15.72 (a) $\sum \vec{\mathbf{F}} = \dfrac{-2Ty}{L}\hat{\mathbf{j}}$; (b) $\omega = \sqrt{\dfrac{2T}{mL}}$

P15.74 If he encounters washboard bumps at the same frequency as the free vibration, resonance will make the motorcycle bounce a lot. It may bounce so much as to interfere with the rider's control of the machine; $\sim 10^1$ m.

P15.76 (a) See ANS. FIG. P15.76(a); (b) 1.74 N/m ± 6%; (c) See table in P15.76(c); (d) See table in P15.76(d); (e) See ANS. FIG. P15.64(e); (f) 1.82 N/m ± 3%; (g) they agree; (h) 8 grams ± 12% in agreement

P15.78 (a) 5.20 s; (b) 2.60 s; (c) $\dfrac{dA/dt}{A} = \dfrac{1}{2}\dfrac{dE/dt}{E}$

P15.80 See P15.80 for complete solution.

P15.82 If the damping constant is doubled, $b/2m = 120\ \text{s}^{-1}$. In this case, however, $b/2m > \omega_0$ and the system is overdamped. Your design objective is not met because the system does not oscillate.

P15.84 (a) $v = 2\left[\dfrac{Rg(1-\cos\theta)}{M/m + r^2/R^2 + 2}\right]^{1/2}$; (b) $2\pi\left[\dfrac{(M+2m)R^2 + mr^2}{2mgR}\right]^{1/2}$

P15.86 This is exactly the same time interval as for your competitor, so you have no advantage! In fact, you have the disadvantage of the initial capital outlay to bore through the entire Earth!

P15.88 (a) $y_f = -0.110$ m; (b) its period will be longer

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.